

Computer Aided Design (CAD)

Circuit for controlling the concentration of carbon monoxide in an enclosure

Student:

Bălășcău Patricia-Andreea

Coordinator:

Assist. Prof. Fizeșan Raul-Traian, PhD

Contents

1. Theoretical background.....	3
1.1. Project Description and Requirements	3
1.2. Block diagram.....	6
1.2.1. Linear active bridge	6
1.2.2. Non-Inverting Linear Voltage Domain Converter	9
1.2.3. Non-inverting hysteresis comparator	11
1.2.4. LED circuit.....	15
1.2.5. Relay circuit module	16
1.3. Incomplete Circuit	17
2. Implementation in PSpice Environment.....	18
2.1. Selection of Components	18
2.1.1. Resistors	18
2.1.2. Operational amplifiers (OA)	18
2.1.3. MOSFETs and drain resistors	19
2.1.4. Orange LED	20
2.1.5. Relay	21
2.2. Simulations	21
2.2.1. DC sweep analysis	21
2.3.2. Worst Case Analysis.....	30
2.2.3. Monte-Carlo Analysis	31
2.2.4. Temperature Analysis Justification	33
3. Bill Of Materials (BOM)	36
4. Conclusion	38
5. Appendix	39
6. References	40

1. Theoretical background

1.1. Project Description and Requirements

This project involves the design and simulation of an automatic control system that keeps carbon monoxide concentration within safe levels in an enclosed space. The sensor used is resistive and reacts linearly to gas concentration changes within the 600 - 17.000 ppm range. In this interval, its resistance decreases from 200 k Ω to 25 k Ω as gas levels increase.

The system must ensure that the CO concentration remains between 700 - 8.000 ppm. When the upper limit is reached, a fan is activated to introduce fresh air. Once the concentration drops to the lower limit, the fan is turned off.

To interpret the sensor's resistance variation, a voltage conversion is applied so that the output signal ranges between 2V and 12V ($[2 \div (V_{CC} - 2V)]$), considering a supply voltage $V_{CC} = 14V$. A hysteresis comparator is used to manage the fan switching logic via a relay, which is modeled with a resistor in the simulation. The fan's status is visually indicated using an orange LED, which turns on when the fan is active.

This configuration ensures reliable operation by converting sensor readings into a clear control signal, with hysteresis helping to avoid unstable switching behavior.

indoor carbon monoxide concentration [ppm]	the measurement range of the sensor [ppm]	Sensor resistance [k Ω]	Vcc [V]	LED Colour
700 – 8.000	600 – 17.000	200 - 25	14	Orange

Table (1). Project Parameter Table

To establish the correct behavior of the control system, the relationship between gas concentration (in ppm) and the sensor's resistance was first modeled using a linear interpolation. Since the sensor's resistance varies linearly between 200 k Ω and 25 k Ω over the range of 600 to 17.000 ppm, the slope of the line was computed in formula (1) and used to derive a linear equation that expresses resistance (R_S) as a function of gas concentration, formulas (2) and (3). Using this equation, the resistance values corresponding to the lower and upper control limits (700 ppm and 8000 ppm) were calculated, shown in formulas (4) and (5).

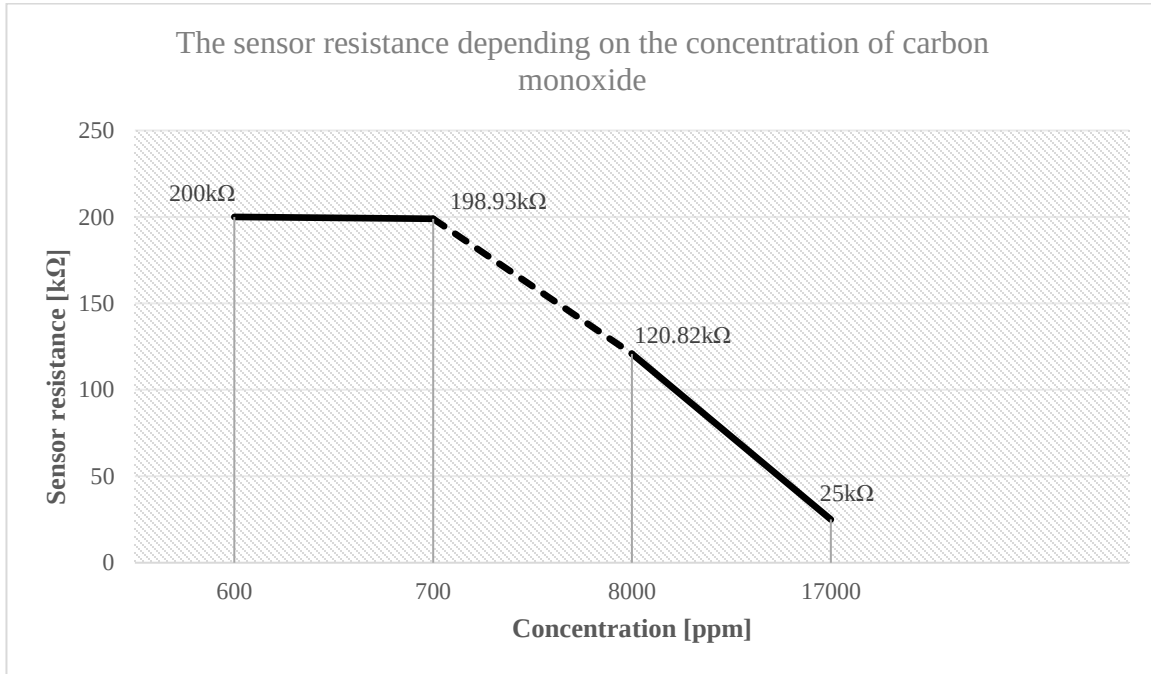


Table (2). The variation of resistance depending on concentration

For each interval, a linear equation was derived from the graph by applying the general form of a straight line and computing the slope between two given data points:

$$p = \frac{y_2 - y_1}{x_2 - x_1} = \frac{25 - 200}{17.000 - 600} = -0.0107 \quad (1)$$

$$y(x) = y_1 + p \cdot (x - x_1) \quad (2)$$

$$R = 200 + p \cdot \Delta ppm \quad (3)$$

For a concentration of 700 [ppm], the resistance value is:

$$R[ppm = 700] = 200 + (-0.0107) \cdot (700 - 600) = 200 - 0.0107 \cdot 100 = 198.93 \text{ [kΩ]} \quad (4)$$

For a concentration of 8.000 [ppm], the resistance value is:

$$R[ppm = 8.000] = 200 + (-0.0107) \cdot (8.000 - 600) = 200 - 0.0107 \cdot 7.400 = 120.82 \text{ [kΩ]} \quad (5)$$

The threshold voltages V_{ThL} and V_{ThH} were extracted from the V_{out2} transfer curve by performing linear interpolation between significant slope points, as indicated in formula (6), based on the variation of sensor resistance R_S , serving as reference thresholds for hysteresis control in the system.

While the calculations and resulting values are identical for both outputs, V_{out2} is analyzed with respect to R_S , and $V_{comparator}$ (V_{out3}) is plotted against CO concentration (ppm), reflecting the same data in different representations.

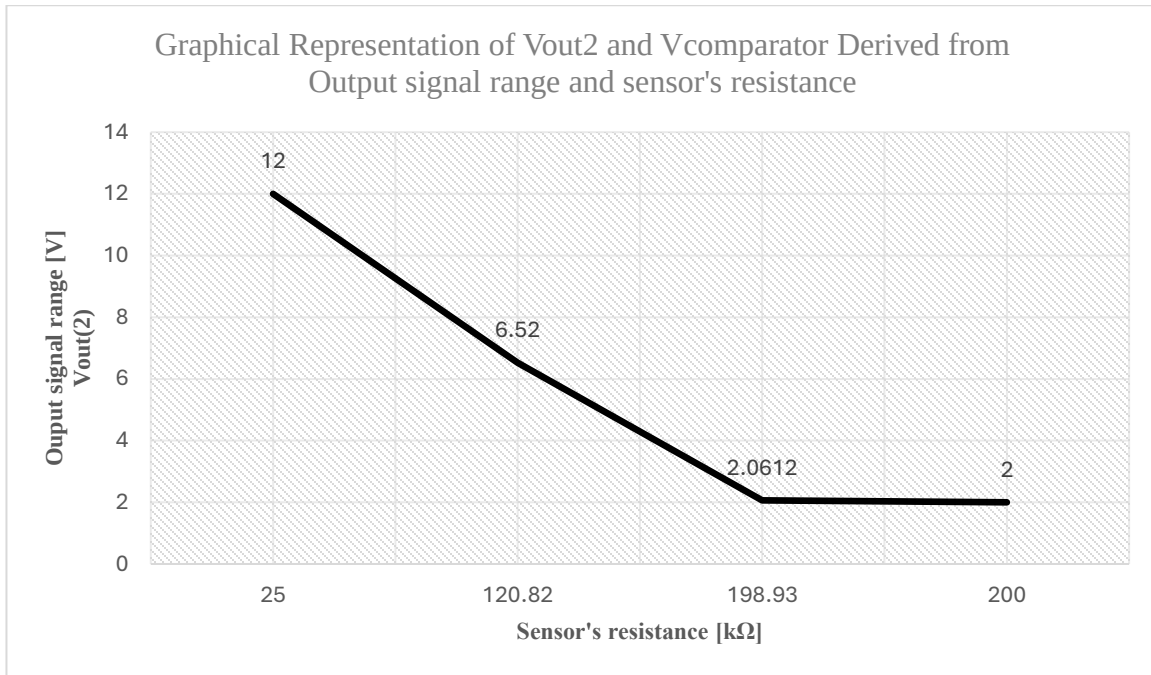


Table (3). The variation of V_{out2} (V_{ThL} and V_{ThH}) depending on sensor's resistance

From the graph, the same corresponding formulas for slope and threshold voltages are presented below:

$$p = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2 - 12}{200 - 25} = -0.057 \quad (6)$$

For a resistance of 120.82 [kΩ], the high threshold value is:

$$V_{out2}(120.82 \text{ k}\Omega) = V_{out2_{max}} + p \cdot (120.82 - R_{s_{min}}) = 12 - 0.057 \cdot (120.82 - 25) = 6.52 \text{ [V]} \quad (7)$$

For a resistance of 198.93 [kΩ], the low threshold value is:

$$V_{out2}(198.93 \text{ k}\Omega) = V_{out2_{min}} + p \cdot (198.93 - R_{s_{max}}) = 2 - 0.057 \cdot (198.93 - 200) = 2.0612 [V] \quad (8)$$

The calculations below are the first step in understanding how the sensor works. By finding the link between resistance, CO concentration, and V_{out2} , they help define the threshold values and support the next stages of the project. The previous results are displayed in table (4).

Quantity				
Concentration CO [ppm]	600	700	8.000	17.000
Sensor's resistance [kΩ]	200	198.93	120.82	25
Output (Vout2) signal range [V]	2	2.0612	6.52	12

Table (4). Final Values

1.2. Block diagram

Figure (1) shows a block diagram of the circuit to help with understanding. The complete circuit will be shown in the next sections.

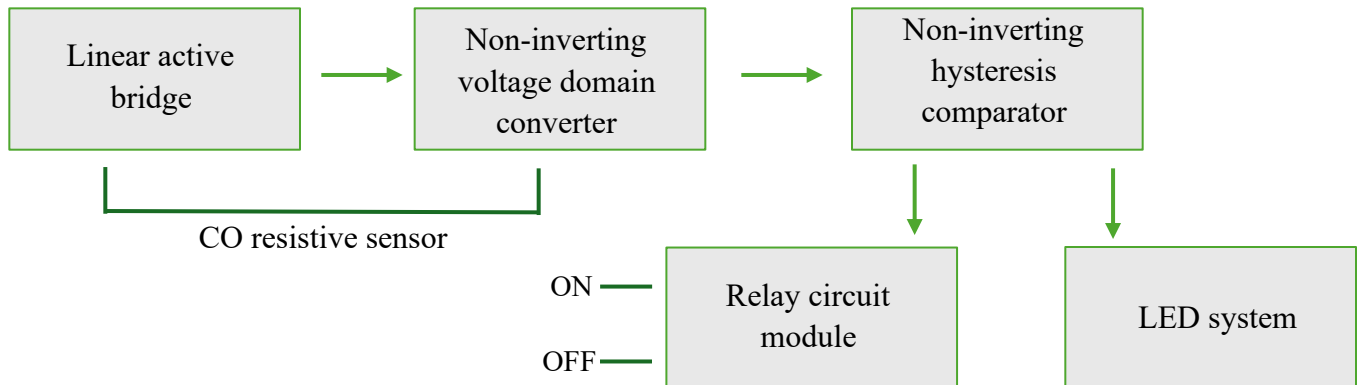


Figure (1). Block Diagram

1.2.1. Linear active bridge

The active bridge is the sensor component that converts the changing resistance into a linear voltage variation. Figure (2) shows the proposed design for this block.

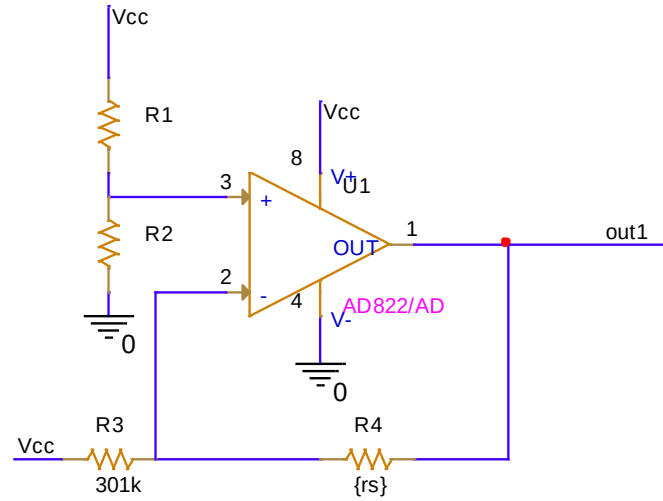


Figure (2). Linear Active Bridge

The linear active bridge uses operational amplifiers with negative feedback to ensure stability and linearity of the output signal. The negative feedback forces the op-amp to adjust its output so that the voltage difference between its inputs remains close to zero.

$$VD = V^+ - V^- \quad (9)$$

$$VD = 0 \Rightarrow V^+ = V^- \quad (10)$$

V^+ is first calculated using the voltage domain equation, as in the formula (11). Then, V^- is found by applying the superposition principle, as shown in formula (12).

$$V^+ = V_{cc} \cdot \frac{R_1}{R_1 + R_2} \quad (11)$$

$$V^- = \frac{R_4}{R_4 + R_3} \cdot V_{cc} + \frac{R_3}{R_3 + R_4} \cdot V_{out1} \quad (12)$$

$$\begin{aligned}
 \frac{R_1}{R_1 + R_2} \cdot V_{cc} &= \frac{R_4}{R_4 + R_3} \cdot V_{cc} + \frac{R_3}{R_3 + R_4} \cdot V_{out1} \Rightarrow \frac{R_3}{R_3 + R_4} \cdot V_{out1} = \left(\frac{R_1}{R_1 + R_2} - \frac{R_4}{R_4 + R_3} \right) \cdot V_{cc} \Rightarrow V_{out1} \\
 &= \frac{R_4 + R_3}{R_3} \cdot \left(\frac{R_1}{R_1 + R_2} - \frac{R_4}{R_4 + R_3} \right) \cdot V_{cc} = \left(\frac{R_4 R_1 + R_3 R_1 - R_4 R_1 - R_4 R_2}{R_3 (R_1 + R_2)} \right) \cdot V_{cc} \\
 &= \left(\frac{R_3 R_1 - R_4 R_2}{R_3 (R_1 + R_2)} \right) \cdot V_{cc} \Rightarrow V_{out1} = \left(\frac{R_1}{R_1 + R_2} - R_4 \cdot \frac{R_2}{R_3 (R_1 + R_2)} \right) \cdot V_{cc}
 \end{aligned}
 \tag{13}$$

We can assume that R_1, R_2, R_3 are equal in order to minimize the equation. So, we denote their common value as "a":

$$V_{out1} = \left(\frac{a}{2 \cdot a} - R_4 \cdot \frac{a}{a \cdot 2 \cdot a} \right) \cdot V_{cc} \Rightarrow V_{out1} = \left(\frac{1}{2} - R_4 \cdot \frac{1}{2a} \right) \cdot 14 = 7 - 7 \cdot \frac{R_4}{a} \tag{14}$$

Since the operational amplifier operates with a single supply, the output voltage V_{out1} must remain above 0V (ground potential). This implies that the ratio $\frac{R_4}{a}$ should stay below 1 for any value of $R_4 = R_S$. In this design, the value for resistor "a" was selected, about 301k Ω ($> 200 \text{ k}\Omega$) as it fulfills this condition and is part of the E96 series (with 1% tolerance).

$$V_{out1} > 0 \Rightarrow 7 - 7 \cdot \frac{R_4}{a} > 0 \Rightarrow \frac{R_4}{a} < 1 \Rightarrow a > R_{4max} \tag{15}$$

$$V_{out1 min} = 7 - 7 \cdot \frac{R_{4max}}{301k\Omega} = 7 - 7 \cdot \frac{200k\Omega}{301k\Omega} = 2.348 [V] \tag{16}$$

$$V_{out1 max} = 7 - 7 \cdot \frac{R_{4min}}{301k\Omega} = 7 - 7 \cdot \frac{25k\Omega}{301k\Omega} = 6.418 [V] \tag{17}$$

1.2.2. Non-Inverting Linear Voltage Domain Converter

The voltage domain converter is a non-hysteresis block that linearly shifts the input signal into the [2V, 12V] range, ensuring that the sensor's output voltage stays within this range, without adding any threshold offsets. This stage takes V_{out1} as its input, along with a reference voltage (V_{Ref}) that helps set the correct output voltage swing. The corresponding circuit is shown in Figure (3).

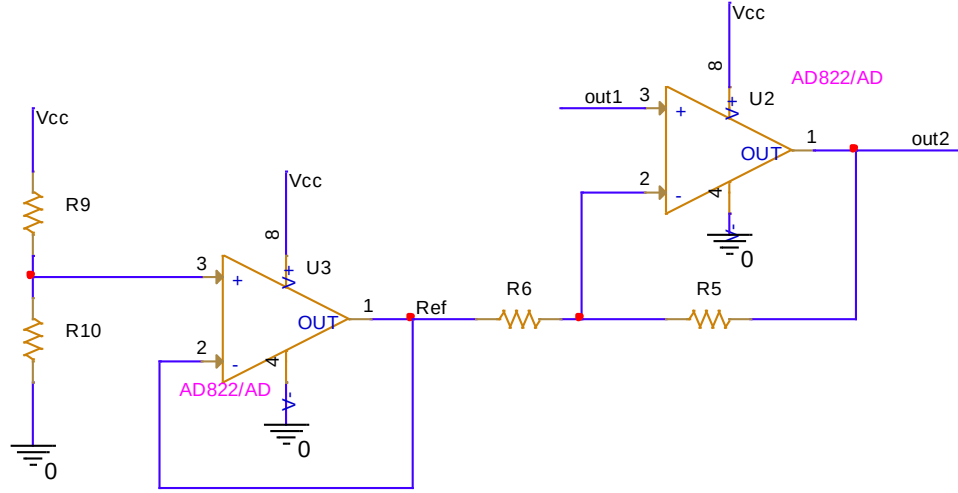


Figure (3). Non-Inverting Voltage Domain Converter

Formula (10) remains applicable for this stage. Equations (18) and (19) define the voltages V^+ and V^- in their final form (see [Electronic Devices, C10 slide 9](#)), respectively:

$$V_{out1} = \frac{R_5}{R_5 + R_6} \cdot V_{Ref} + \frac{R_6}{R_6 + R_5} \cdot V_{out2} \quad (18)$$

$$V_{out2} = \left(1 + \frac{R_5}{R_6}\right) \cdot V_{out1} - \frac{R_5}{R_6} \cdot V_{Ref} \quad (19)$$

Since V^+ represents the non-inverting voltage domain, the minimum and maximum values of V_{out2} are directly proportional to the corresponding minimum and maximum values of V_{out1} , represented in the table (5), which were calculated in the previous circuit block, formulas (16) and (17).

Voltage (max,min)	Value [V]	
V_{out2}	12	2
V_{out1}	6.418	2.348

Table 5. Output voltages for Non-Inverting Voltage Domain

To simplify the equation and determine the resistor values, we apply the same method using the variable 'a'.

Therefore, the following formula applies:

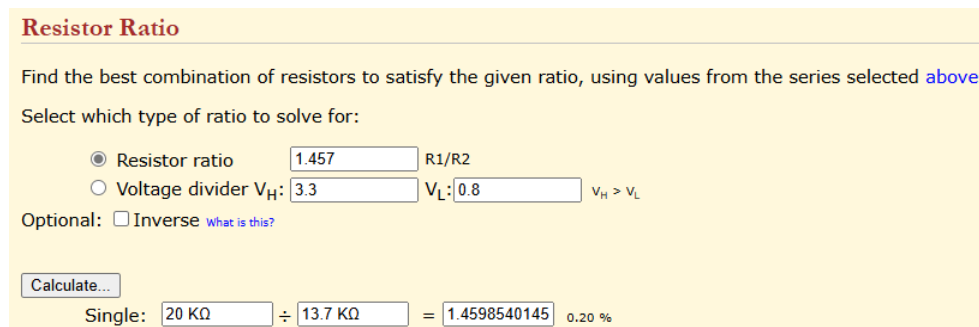
$$\begin{cases} V_{out2\ min} = (1 + a) \cdot V_{out1\ min} - a \cdot V_{ref} \\ V_{out2\ max} = (1 + a) \cdot V_{out1\ max} - a \cdot V_{ref} \end{cases} \Leftrightarrow \begin{cases} 2 = (1 + a) \cdot 2.348 - a \cdot V_{ref} \\ 12 = (1 + a) \cdot 6.418 - a \cdot V_{ref} \end{cases} \quad (20)$$

After simplification, the system reduces to the final form:

$$10 = (1 + a) \cdot 4.07 \Rightarrow (1 + a) = 2.457 \Rightarrow a = 1.457 \quad (21)$$

$$a = \frac{R_5}{R_6} = 1.457 \Rightarrow \begin{cases} R_5 = 20\ k\Omega \\ R_6 = 13.7\ k\Omega \end{cases} \quad (22)$$

Subsequently, the values of resistors R6 and R7 were selected based on the closest standard values obtained using an online resistor value calculator [6], as shown in Figure (4). (see [Resistance Calculator](#)).



Resistor Ratio

Find the best combination of resistors to satisfy the given ratio, using values from the series selected [above](#).

Select which type of ratio to solve for:

☒ Resistor ratio R1/R2

☐ Voltage divider V_H : V_L : $V_H > V_L$

Optional: ☐ Inverse [What is this?](#)

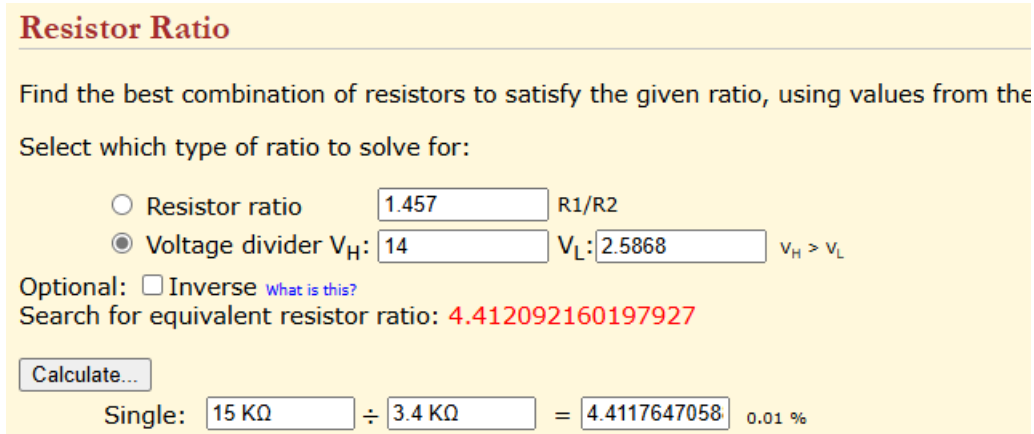
Single: ÷ = 0.20 %

Figure 4. Finding the values for R5 and R6

As a final step, 'a' in equation (20) was expressed as $\frac{R_5}{R_6}$, and the values of R_7 and R_8 were determined using the same online calculator, after computing V_{Ref} , presented below:

$$V_{Ref} = \frac{V_{out2\ min} - (V_{out2\ max} - V_{out1\ max})}{-1.457} = \frac{2 - 5.769}{-1.457} \Rightarrow V_{Ref} = 2.5868 \Rightarrow \begin{cases} R_7 = 15\ k\Omega \\ R_8 = 3.4\ k\Omega \end{cases} \quad (23)$$

In the end, given the known value of V_{Ref} in the formula (23), the resistor values R_7 and R_8 are determined using the voltage divider formula, calculated online.



Resistor Ratio

Find the best combination of resistors to satisfy the given ratio, using values from the

Select which type of ratio to solve for:

☐ Resistor ratio R1/R2

☒ Voltage divider V_H : V_L : $V_H > V_L$

Optional: ☐ Inverse [What is this?](#)

Search for equivalent resistor ratio: **4.412092160197927**

Single: ÷ = 0.01 %

Figure (5). Finding the values for R_7 and R_8

Therefore, the resistive sensor for carbon monoxide detection has been designed and dimensioned.

1.2.3. Non-inverting hysteresis comparator

This stage detects when the input voltage (V_{out2}) crosses defined thresholds and generates a digital output (V_{out3}). It uses an AD822 operational amplifier configured as a non-inverting comparator with hysteresis, achieved through positive feedback via resistor R_9 .

A reference voltage (V_{ref}), set by a resistor divider (R_{11} , R_{12}), is applied to the inverting input. When the input rises above the upper threshold, the output switches to a high level (14V). When it falls below the lower threshold, the output returns to a low level (0V). This behavior controls the fan: it activates when CO concentration exceeds the upper limit and deactivates when the level drops below the lower limit, ensuring stable and efficient operation.

The circuit for this stage can be viewed in figure (6).

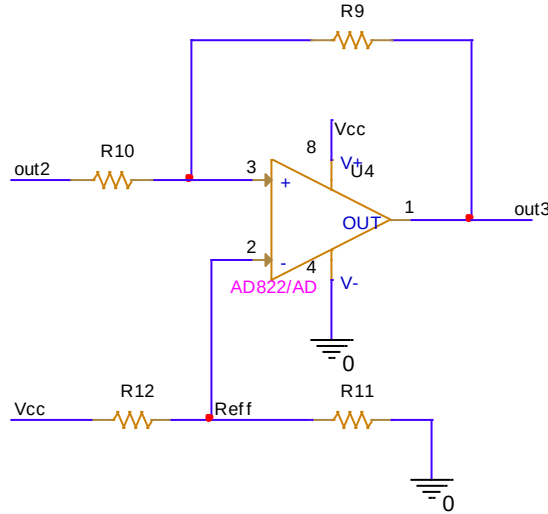


Figure (6). Non-inverting Hysteresis Comparator

The asymmetric threshold values (V_{ThL} and V_{ThH}) were calculated using equations (7) and (8), based on their relationship with output signal voltage (V_{out2}) and the sensor's variable resistance, as summarized in Table [4].

To determine the appropriate threshold voltages, it is important to consider that V_{out3} operates in only two stable states: $V_{out3L} = 0V$ and $V_{out3H} = 14V$.

Based on the formulas for the non-inverting input voltage (V^+) and the inverting input voltage (V^-), the expression for the input voltage difference can be derived, as explained in (see [Electronic Devices C7 slide 17](#)).

$$V^+ = \frac{R_9}{R_9 + R_{10}} \cdot V_{out2} + \frac{R_{10}}{R_{10} + R_9} \cdot V_{out3} \quad (24)$$

$$V^- = V_{Reff} = \frac{R_{11}}{R_{11} + R_{12}} \cdot V_{Cc} \quad (25)$$

Using the formula (9), the expression for the comparator input difference can be derived:

$$\frac{R_9}{R_9+R_{10}} \cdot V_{out2} + \frac{R_{10}}{R_{10}+R_9} \cdot V_{out3} - V_{Reff} = 0 \quad (26)$$

Within the hysteresis mechanism, the threshold voltage V_{Th} is defined by the input voltage V_{in} , denoted here as V_{out2} , specified in the formula (27):

$$V_{Th} = V_{in} = V_{out2} | V_D = 0 \quad (27)$$

$$\frac{R_9}{R_9+R_{10}} \cdot V_{Th} + \frac{R_{10}}{R_{10}+R_9} \cdot V_{out3} - V_{Reff} = 0 \Rightarrow \frac{R_9}{R_9+R_{10}} \cdot V_{Th} = V_{Reff} - \frac{R_{10}}{R_{10}+R_9} \cdot V_{out3} \quad (28)$$

The formula for V_{Th} is shown in formula (29):

$$V_{Th} = (V_{Reff} - \frac{R_{10}}{R_{10}+R_9} \cdot V_{out3}) \cdot \frac{R_9+R_{10}}{R_9} \Rightarrow V_{Th} = (1 + \frac{R_{10}}{R_9}) \cdot V_{Reff} - \frac{R_{10}}{R_9} \cdot V_{out3} \quad (29)$$

To continue, V_{Th} can be replaced in formula (29) with V_{ThL} and V_{ThH} which were previously calculated and replace the values for V_{out3} with: $V_{out3} \in \{0V, 14V\}$. Since the comparator uses a non-inverting hysteresis configuration, V_{Th} and V_{out3} are inversely proportional: a higher threshold V_{ThH} corresponds to the minimal output value (0V), and a lower threshold V_{ThL} corresponds to the maximal output value (14V).

The expression of V_{ThH} is shown in the next formula (30):

$$V_{ThH} = (1 + \frac{R_{10}}{R_9}) \cdot V_{Reff} - \frac{R_{10}}{R_9} \cdot V_{out3 \min} \quad (30)$$

And the expression of V_{ThL} is shown in the next formula (31):

$$V_{ThL} = (1 + \frac{R_{10}}{R_9}) \cdot V_{Reff} - \frac{R_{10}}{R_9} \cdot V_{out3 \max} \quad (31)$$

Now, to determine the appropriate resistor values to be selected from the E96 series, the threshold voltage equations are simplified by substituting the resistor ratio $\frac{R_{10}}{R_9}$ with the variable 'a', resulting in the form shown in formula (32).

$$\begin{cases} 6.52 = (1 + a) \cdot V_{Ref} - a \cdot 0 \\ 2.0612 = (1 + a) \cdot V_{Ref} - a \cdot 14 \end{cases} \Rightarrow 4.4588 = 14 \cdot a \Rightarrow a = 0.3184 \Rightarrow \frac{R_{10}}{R_9} = 0.3184 \quad (32)$$

$$\Rightarrow \begin{cases} R_{10} = 107 \text{ k}\Omega \\ R_9 = 34 \text{ k}\Omega \end{cases}$$

Using the ratio resistance calculator mentioned in figure (7).

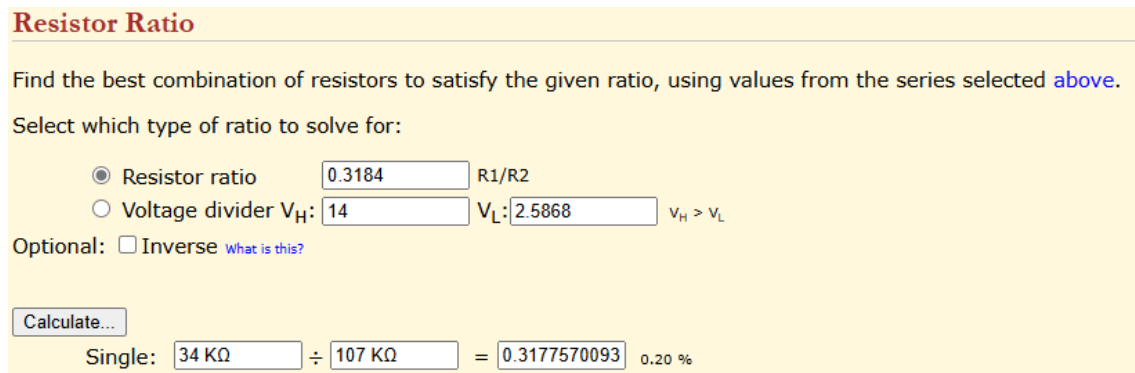


Figure (7). Finding the values for R10 and R9

By using formulas (32) and (25), the ratio $\frac{R_{11}}{R_{11}+R_{12}}$ can be found:

$$V_{Ref} = \frac{V_{out2 \max}}{1 + a} = \frac{6.52}{1.3184} = 4.94538 \quad (33)$$

$$V_{Ref} = \frac{R_{11}}{R_{11}+R_{12}} \cdot V_{Cc} \Rightarrow 4.94538 = \frac{R_{11}}{R_{11}+R_{12}} \cdot 14 \Rightarrow \begin{cases} R_{11} = 10.7 \text{ k}\Omega \\ R_{12} = 19.6 \text{ k}\Omega \end{cases} \quad (34)$$

The E96 resistor values that meet the equation are given in the figure (8):

Resistor Ratio

Find the best combination of resistors to satisfy the given ratio, using values from the series selected [above](#).

Select which type of ratio to solve for:

☐ Resistor ratio R1/R2

☒ Voltage divider V_H : V_L : $V_H > V_L$

Optional: ☐ Inverse [What is this?](#)

Search for equivalent resistor ratio: **1.8309250249728026**

Single: ÷ = 0.05 %

Figure (8). Finding the values for R11 and R12

In summary, the hysteresis comparator stage has been fully defined and dimensioned in the preceding section.

1.2.4. LED circuit

In the CO monitoring system, the LED signals turns on when the comparator output connected to the CO sensor is HIGH, indicating that the CO level has passed the set limit. A key design point was to ensure the LED receives the proper current when lit. To do this, an N-type MOSFET was used as a current switch.

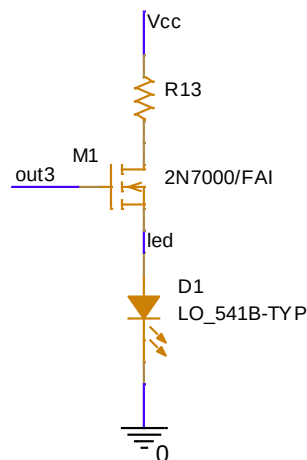


Figure (9). LED Circuit

The circuit was sized based on practical testing and designed according to the datasheet specifications, considering a forward current of 20mA, a forward voltage of 2V for the LED, and a supply voltage of 5V. Using Ohm's law, the LED resistor was calculated to be 150Ω, as shown in the following formula (35).

$$R_{LED} = \frac{V_{source} - V_{LED}}{I_{LED}} = \frac{5-2V}{20mA} = \frac{3V}{0.02A} = 150\Omega \quad (35)$$

1.2.5. Relay circuit module

Electrically operated switches like relays commonly use an electromagnet. A specific requirement for this work was to model the relay as a resistive load, where the resistance is effectively the coil's inductive reactance (found in the datasheet [10]). The relay's activation voltage from the datasheet needs a certain current to work properly. This current was found using Ohm's Law, formula (36), and supplied by an N-type MOSFET to keep the right voltage drop. The circuit was then finalized based on practical testing.

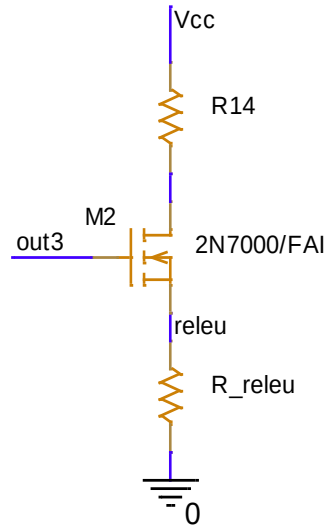


Figure (10). Relay Circuit

Using Ohm's Law, as in the equation above (35), the required current for a 5v relay was calculated as shown in Equation (36).

$$I_{Relay} = \frac{V_{Relay}}{R_{Relay}} = \frac{5V}{125\Omega} = 0.04A = 40mA \quad (36)$$

1.3. Incomplete Circuit

The figure below, figure (11), shows the interconnected blocks of the circuit as described earlier. This schematic is incomplete and does not include component values or specific details, which will be provided later when the full circuit is presented.

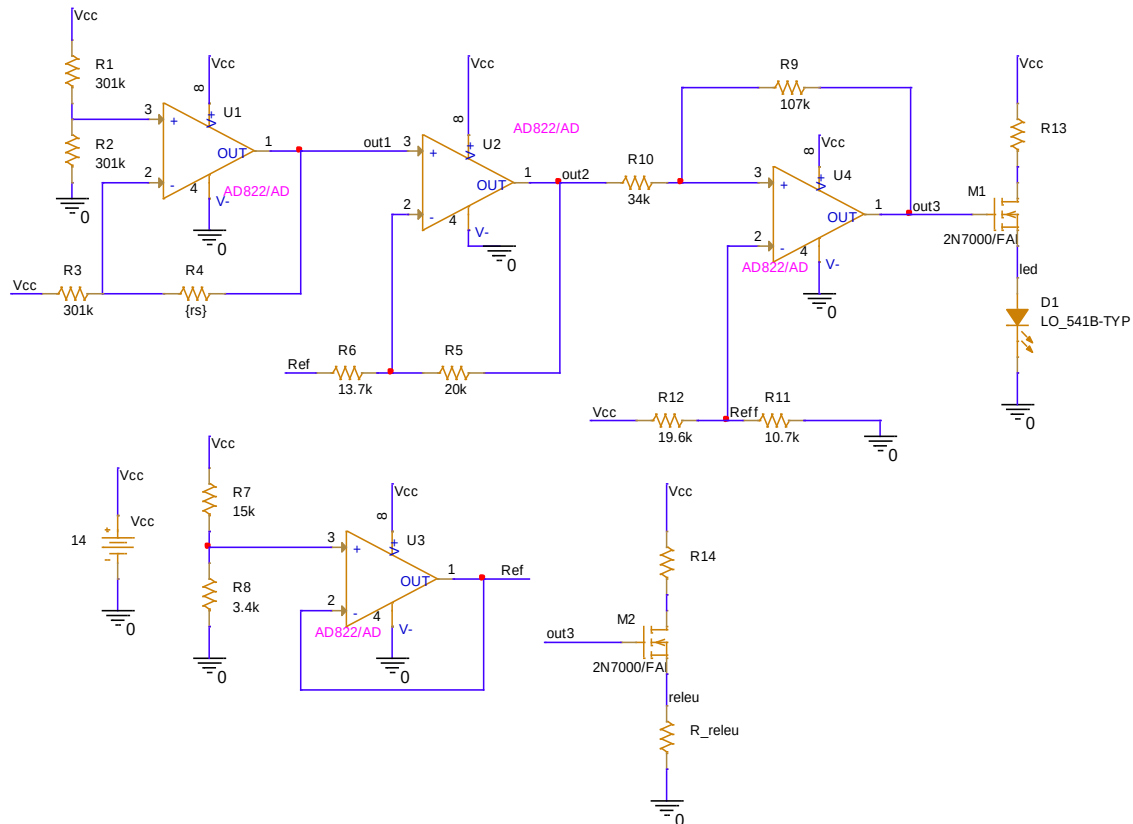


Figure (11). Incomplete Circuit

2. Implementation in PSpice Environment

2.1. Selection of Components

2.1.1. Resistors

All resistors used in this project are part of the E96 standard series (Figure (12)), chosen for their 1% tolerance, which was essential to ensure circuit accuracy. They are through-hole components, each with a nominal power rating of $\frac{1}{4}W$.

1% Resistor Table (E96)											
100	102	105	107	110	113	115	118	121	124	127	130
133	137	140	143	147	150	154	158	162	165	169	174
178	182	187	191	196	200	205	210	215	221	226	232
237	243	249	255	261	267	274	280	287	294	301	309
316	324	332	340	348	357	365	374	383	392	402	412
422	432	442	453	464	475	487	499	511	523	536	549
562	576	590	604	619	634	649	665	681	698	715	732
750	768	787	806	825	845	866	887	909	931	953	976
Standard Values											

Figure (12). Values of E96 resistance series.

2.1.2. Operational amplifiers (OA)

Key selection criteria for the operational amplifier included single 14V supply operation and a rail-to-rail output for the hysteresis comparator's voltage accuracy. The AD822 from Analog Devices met these requirements, with its datasheet [3] verifying single-supply capability (5V-30V) and rail-to-rail output (see Figure (13)).



Single-Supply, Rail-to-Rail Low Power FET-Input Op Amp

Data Sheet

AD822

FEATURES

True single-supply operation

Output swings rail-to-rail

Input voltage range extends below ground

Single-supply capability from 5 V to 30 V

Dual-supply capability from ± 2.5 V to ± 15 V

High load drive

Capacitive load drive of 350 pF, $G = +1$

Minimum output current of 15 mA

CONNECTION DIAGRAM

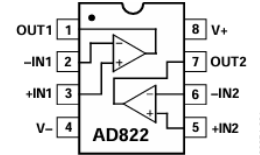


Figure 1. 8-Lead PDIP (N Suffix);
8-Lead MSOP (RM Suffix);
and 8-Lead SOIC_N (R Suffix)

Figure (13). OA AD822 datasheet

2.1.3. MOSFETs and drain resistors

Two N-channel MOSFETs (2N7000) were used to supply enough current to the relay and the LED, since the comparator cannot drive them directly. These transistors allow current to flow when a “HIGH” voltage is applied to the gate. The 2N7000 was chosen due to its 20 V gate-to-source voltage limit [10]. MOSFETs were preferred over BJTs because their gate is insulated by a dielectric, meaning they draw virtually no current, making them more suitable for voltage-driven control.

2N7000, 2N7002, NDS7002A

ABSOLUTE MAXIMUM RATINGS Values are at $T_C = 25^\circ\text{C}$ unless otherwise noted.

Symbol	Parameter	Value			Unit
		2N7000	2N7002	NDS7002A	
V_{DSS}	Drain-to-Source Voltage	60			V
V_{DGR}	Drain-Gate Voltage ($R_{GS} \leq 1$ MW)	60			V
V_{GSS}	Gate-Source Voltage – Continuous	± 20			V
	Gate-Source Voltage – Non Repetitive ($t_p < 50$ ms)	± 40			
I_D	Maximum Drain Current – Continuous	200	115	280	mA
	Maximum Drain Current – Pulsed	500	800	1500	

Figure (14). 2N7000 N-type MOS gate-source maximum voltage (V_{GS})

2.1.4. Orange LED

The selected LED for this project is the LO_541B_TYP, which emits orange light. According to the datasheet, it has a peak emission wavelength of 610 nm at a forward current of 20 mA. The forward voltage typically measures 2V, with a maximum of 2.4V under the same current conditions.

Kennwerte Characteristics ($T_A = 25^\circ\text{C}$)

Bezeichnung Parameter	Symbol Symbol	Werte Values			Einheit Unit
		LA	LO	LY	
Wellenlänge des emittierten Lichtes (typ.) <u>Wavelength at peak emission</u> $I_F = 20\text{ mA}$	λ_{peak}	624	610	594	nm

Durchlassspannung ^{7) Seite 13} <u>Forward voltage</u> ^{7) page 13} $I_F = 20\text{ mA}$	(min.)	V_F	1.8	1.8	1.8	V
	(typ.)	V_F	2.0	2.0	2.0	V
	(max.)	V_F	2.4	2.4	2.4	V

Figure (15). LO_541B_TYP wavelength and forward voltage in datasheet

LA 541B, LO 541B, LY 541B

Bestellinformation Ordering Information

Typ Type	Emissions- farbe Color of Emission	Gehäuse- farbe Color of Package	Partieller Lichtfluss ¹⁾ Seite 13 Partial Flux ¹⁾ page 13 $I_F = 20\text{ mA}$ $E_V [\text{lux}]$	Lichtstrom ²⁾ Seite 13 Luminous Flux ²⁾ page 13 $I_F = 20\text{ mA}$ $\Phi_V (\text{lm})$	Bestellnummer Ordering Code
LA 541B-AWBW-24 LA 541B-BWDW-24 LA 541B-AWDW-24	amber	colorless clear	1120 ... 2800 1800 ... 7100 1120 ... 7100	630 (typ.) 1425 (typ.) 1315 (typ.)	Q65110A2846 Q65110A2847 Q65110A1261
LO 541B-BWDW-24	orange	colorless clear	1800 ... 7100	1425 (typ.)	Q65110A2848
■LY 541B-AWBW-26 ■LY 541B-BWDW-26 ■LY 541B-AWDW-26	yellow	colorless clear	1120 ... 2800 1800 ... 7100 1120 ... 7100	630 (typ.) 1425 (typ.) 1315 (typ.)	Q65110A2849 Q65110A5247 Q65110A3550

Figure (16). LO_541B_TYP color stated in datasheet

For optimal performance and to ensure the LED operates within its specified parameters, a current-limiting resistor must be added in series. Its value was previously calculated and demonstrated using Ohm's Law, in Equation (36), during the design of the LED circuit block.

2.1.5. Relay

In the simulation, the relay was approximated using a $70\ \Omega$ resistor, matching the coil resistance specified in the datasheet. To ensure the required 5V activation voltage across it, a steady current was calculated. This value was obtained by applying formula (35), substituting the parameters provided in the datasheet [11].

2.2. Simulations

This section shows how the circuit works using the defined blocks with given values and tolerances. The next simulations confirm their functionality and the accuracy of the calculations.

2.2.1. DC sweep analysis

For the linear active bridge block, a DC sweep analysis was conducted using a global parameter derived from the threshold values calculated in equations (7) and (8). This simulation serves to validate the circuit's switching behavior near the specified thresholds.

The primary objective of this simulation is to demonstrate the active bridge's linear conversion of changes in sensor resistance (R_s) into the output voltage (V_{out1}). The resulting plot, presented in Figure (17), clearly confirms this linearity and aligns the output voltage range with the previously determined thresholds, $V_{out1\ min}$ and $V_{out1\ max}$, as calculated in Figure (18).

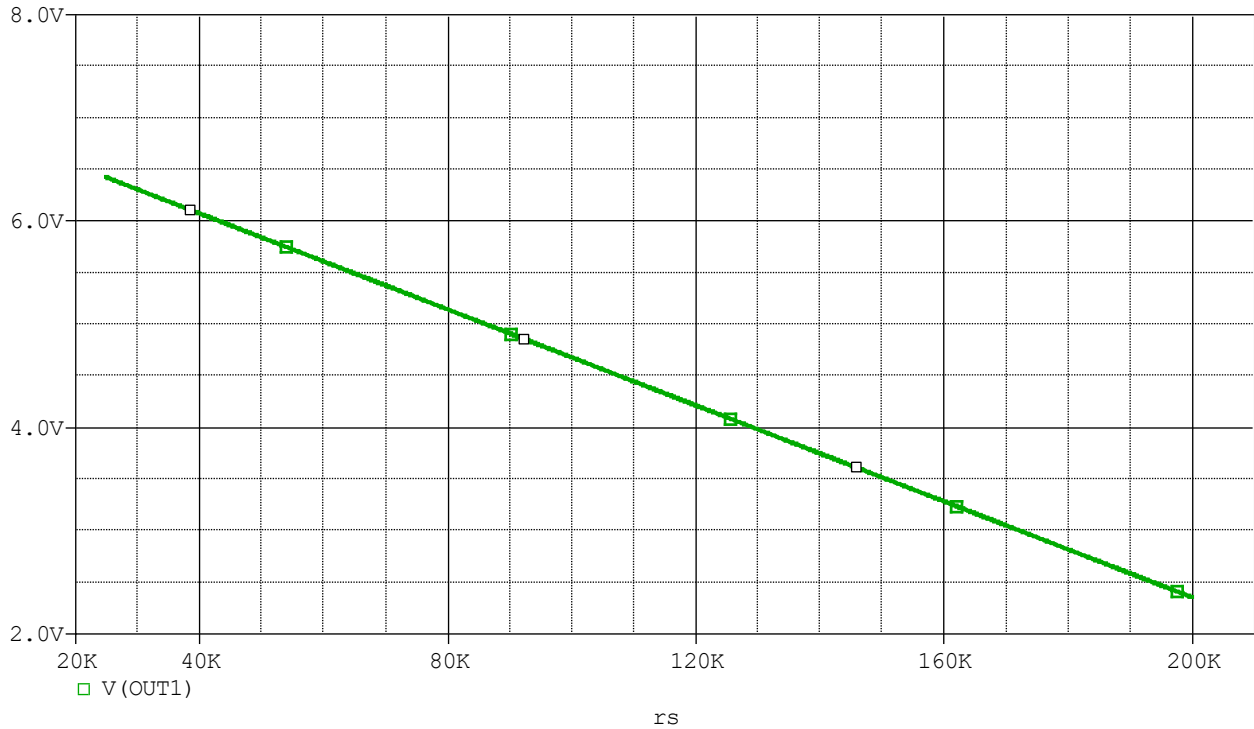


Figure (17). Linear Active Bridge simulation

Trace Color	Trace Name	Y1	Y2
	X Values	25.200K	200.000K
CURSOR 1,2	V(OUT1)	6.4141	2.3490

Figure (18).

After the active bridge, the signal is processed by the non-inverting voltage domain converter stage implemented with U2 (Figure (3)). To evaluate its behavior, a DC sweep analysis was performed by varying the global parameter R_S , simulating changes in sensor resistance.

As shown in Figure (19), this stage linearly maps the resistance range to a corresponding voltage range at V_{out2} . The plot confirms a clear linear relationship, where the output varies from approximately 2V (for $R_S = 200 \text{ k}\Omega$) to 12V (for $R_S = 25 \text{ k}\Omega$), as also highlighted in Figure (20).

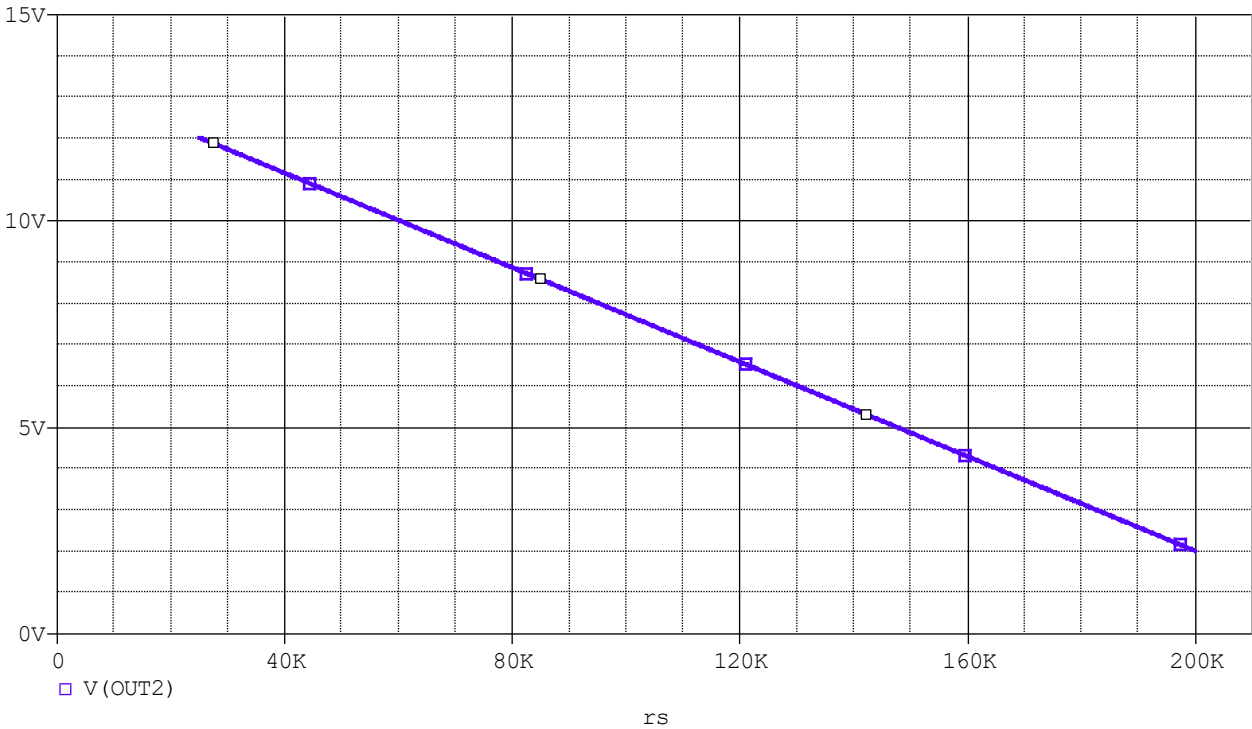


Figure (19). Non-Inverting Voltage Domain Convertor Simulation

Trace Color	Trace Name	Y1	Y2	Y1 - Y2	Y1(Cursor1) - Y2(Cursor2)		10.000	
	X Values	25.200K	200.000K	-174.800K	Y1 - Y1(Cursor1)	Y2 - Y2(Cursor2)	Max Y	Min Y
CURSOR 1,2	V(OUT2)	12.002	2.0013	10.000	0.000	0.000	12.002	2.0013

Figure (20).

The next step was to check how the comparator with hysteresis responds to different sensor resistance values. A DC sweep analysis was performed to observe the output voltage V_{out3} as the sensor resistance R_S changed.

As shown in Figure (21), when R_S is about 200kΩ, V_{out3} remains *HIGH* at around 14V. Once R_S goes below this value, the output quickly switches to *LOW*, close to 0V.

To further verify the hysteresis effect, the simulation was run again with R_S near 25kΩ. Here, the output switched from *HIGH* to *LOW* at this lower resistance threshold, showing clear switching behavior depending on R_S .

These results confirm that the comparator stage gives a stable digital output, switching at specific sensor resistance points, which is typical for a non-inverting comparator with hysteresis.

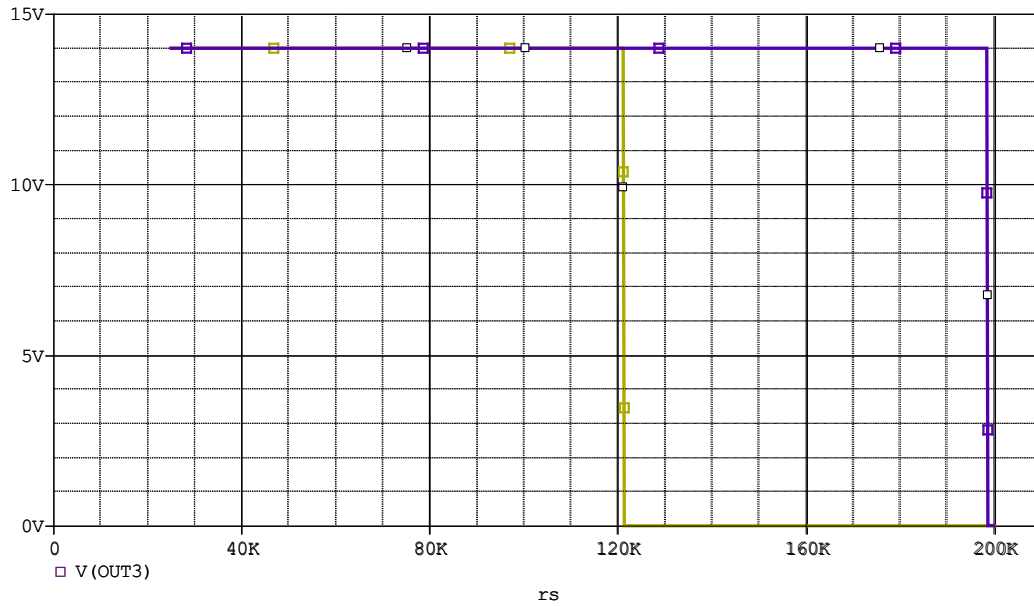


Figure (21). Non-Inverting Hysteresis Comparator Simulation

Figure (21) shows the output voltage behavior when R_s decreases from higher values down to 200 $k\Omega$, highlighting the switching point where the output transitions from *HIGH* to *LOW*.

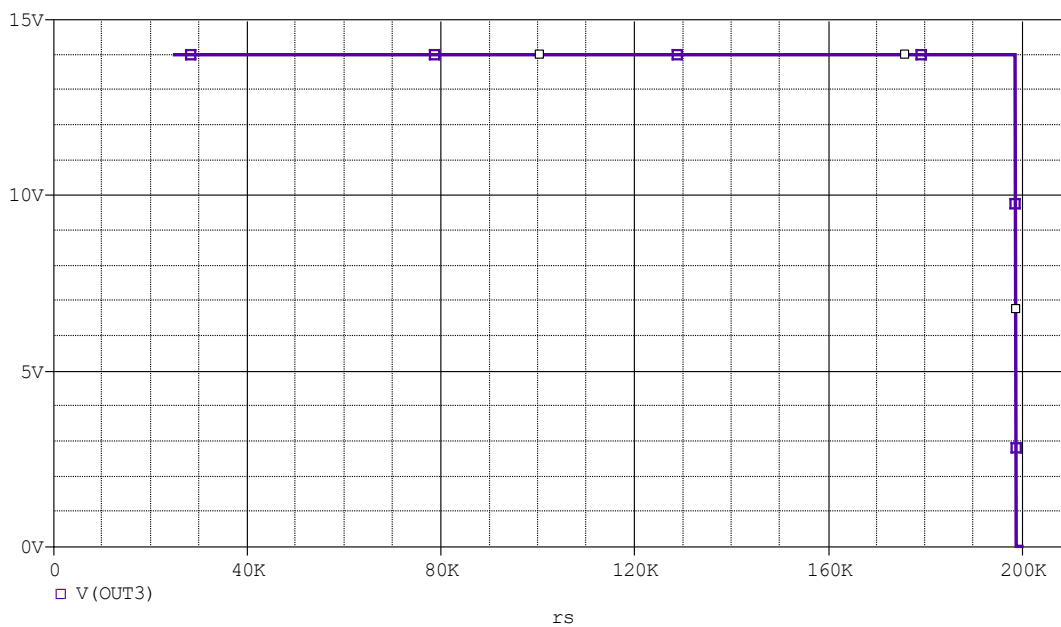


Figure (22). DC sweep of $V(OUT3)$ vs R_s showing output switching from HIGH to LOW at approximately 200 $k\Omega$.

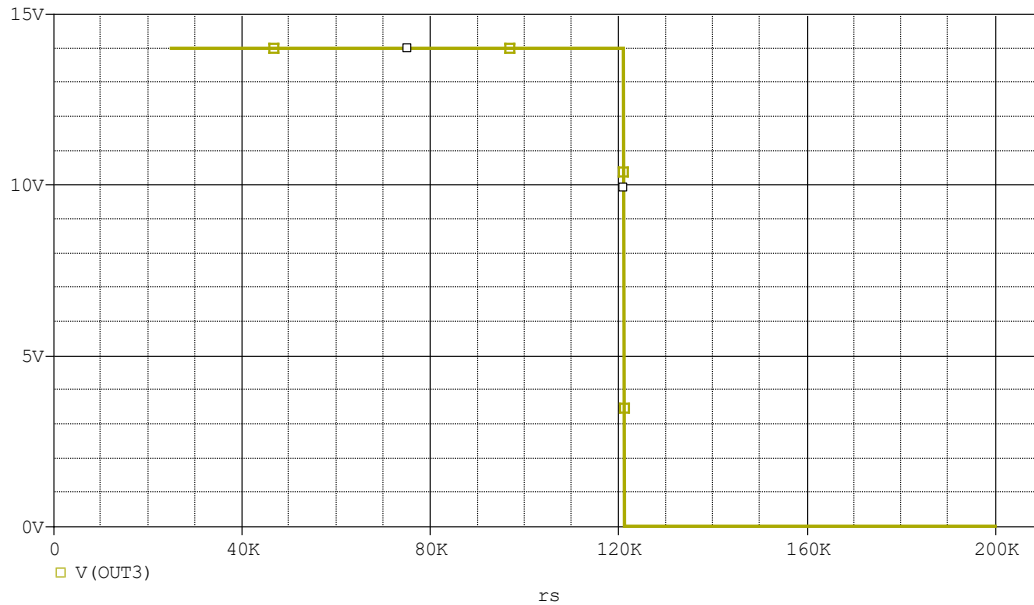


Figure (23). DC sweep of V(OUT3) vs Rs showing output switching from LOW to HIGH at approximately 25kΩ

Figure (23) presents the reverse sweep, where Rs increases from lower values up to around 25kΩ, demonstrating the hysteresis effect as the output switches back from *LOW* to *HIGH*.

To find the right value for resistor R_{13} used with the orange LED, a DC sweep simulation with global parameter (rled) was done by varying the resistance. The goal was to get a current of about 20 mA through the LED, which is the typical value for standard LEDs.

Using the simulation plot and a cursor, the point where the LED current reached 20 mA was identified. A zoomed-in version of the graph is shown in Figure (24). From the plot, the resistor value needed was around 597 Ω (see Figure(26)).

To keep things practical, the closest value from the E96 standard series was chosen. So, R13 was set to 604 Ω, which ensures the LED works properly with a real resistor value.

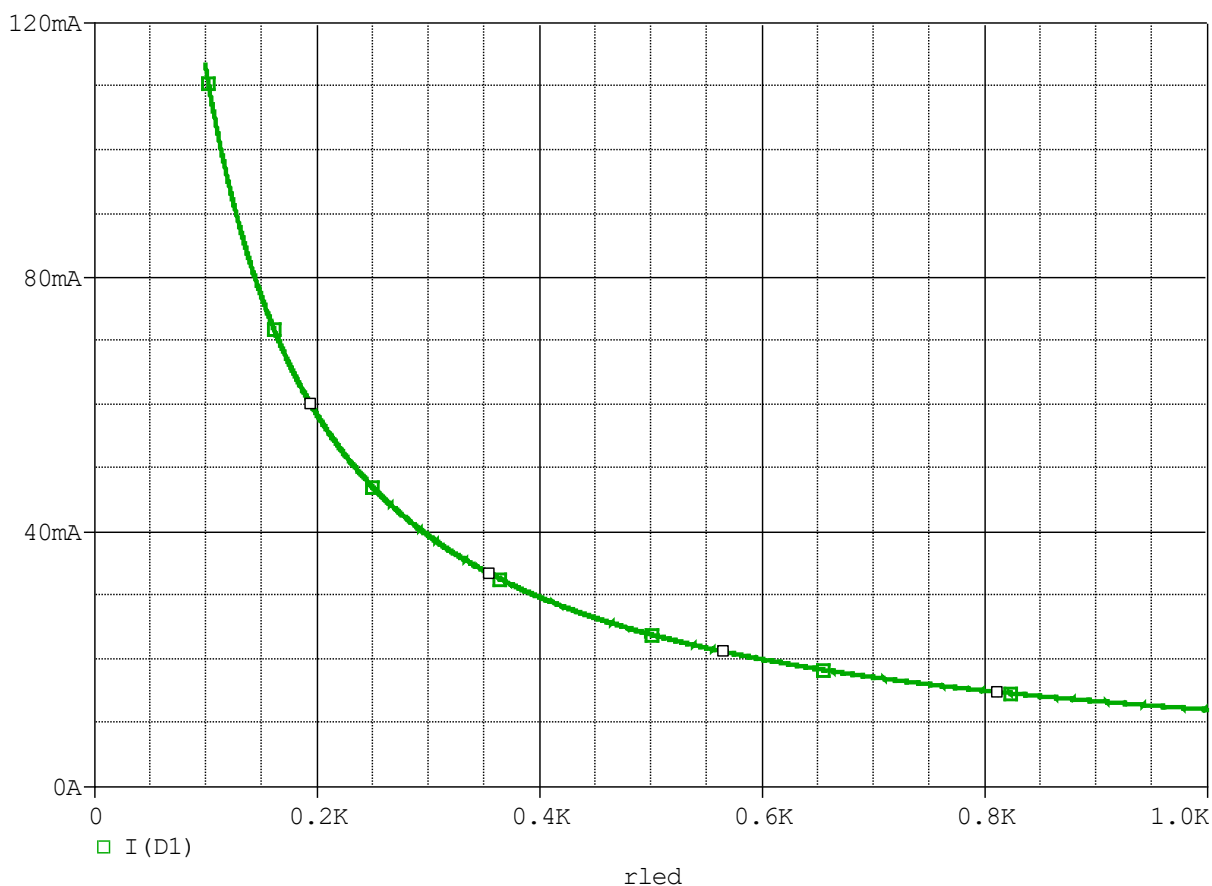


Figure (24). Setting the R13 of the Orange LED

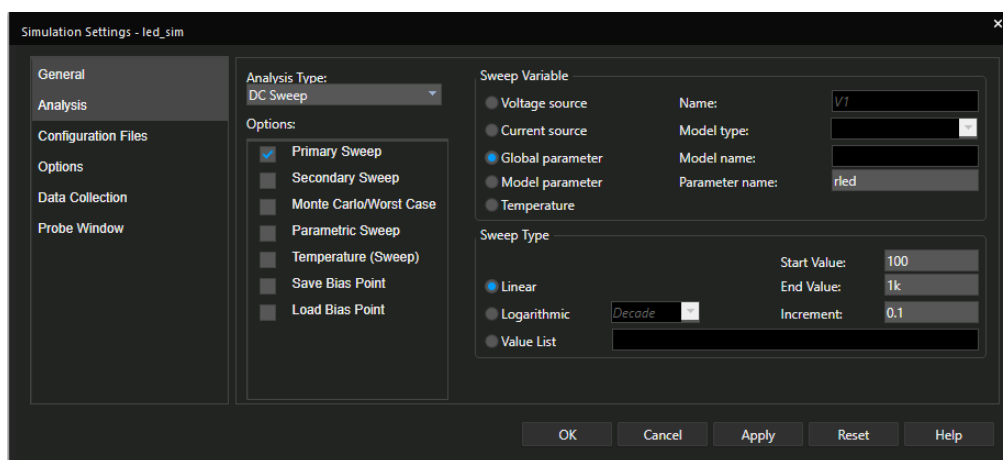


Figure (25). Simulation Profile to find R13

Trace Color	Trace Name	Y1	Y2
	X Values	597.300	9.918
CURSOR 1,2	I(D1)	20.039m	732.620m

Figure (26). Value of R13

The same method used for the LED was applied to the 5V relay. A DC sweep was performed to find the resistor that ensures the required activation current of 40 mA, as specified in the datasheet [].

From the plot, the resistor value that gave this current was around 223 Ω , so the closest E96 standard value, 221 Ω , was chosen, see Figure (29).

Figure (26) shows the point where the relay current reaches 40 mA.

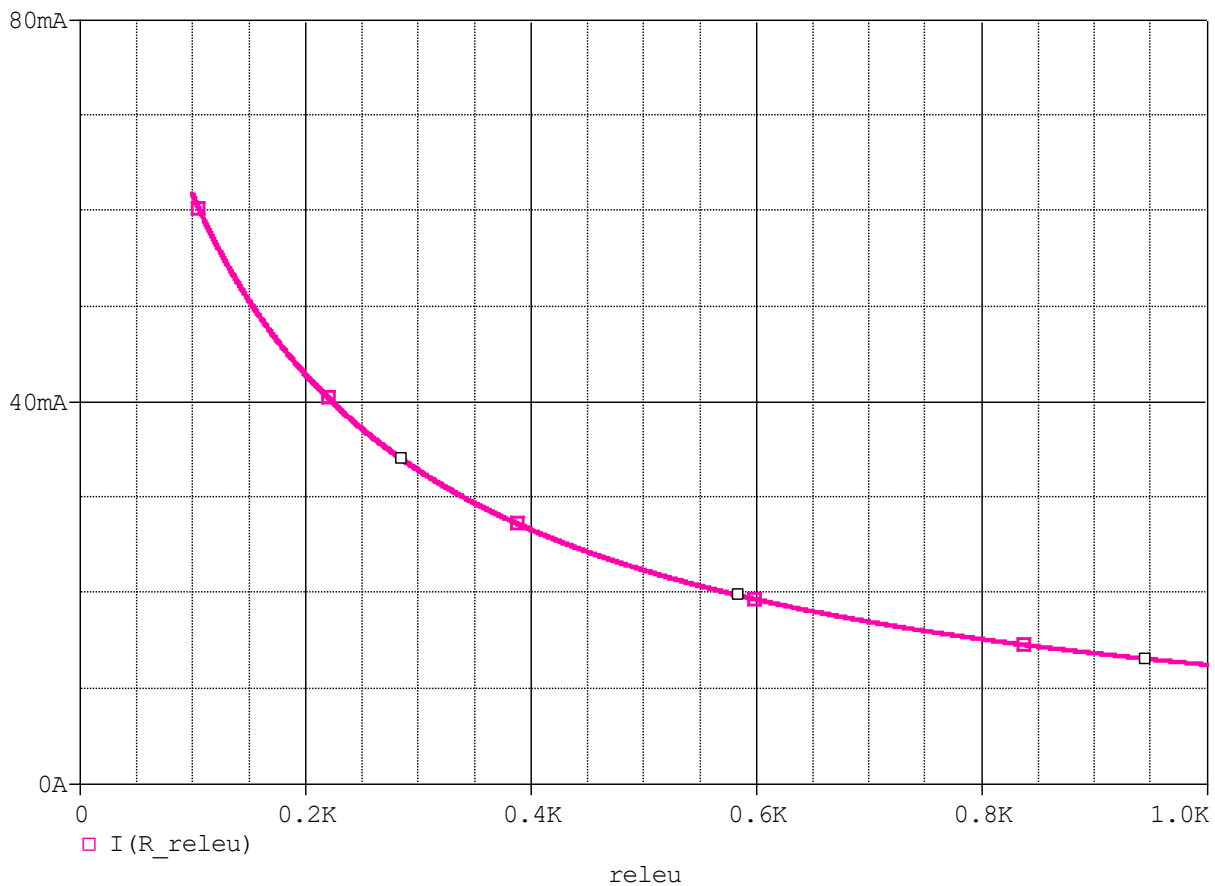


Figure (27). Setting the R14 of the 5V Relay

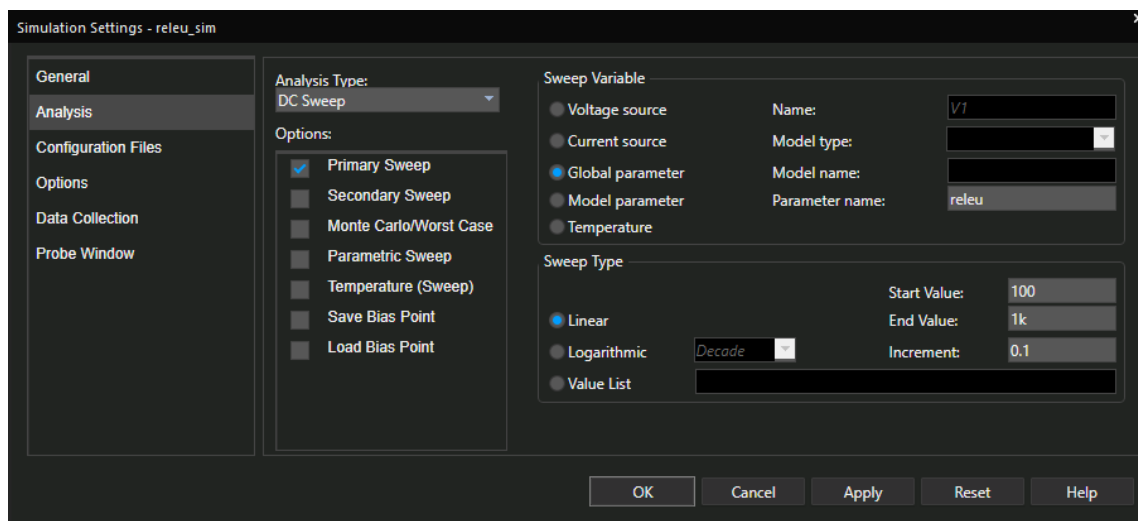


Figure (28). Simulation Profile to find R14

Trace Color	Trace Name	Y1	Y2
	X Values	223.100	223.100
CURSOR 1,2	I(R_releu)	40.061m	40.061m

Figure (29). Value of R14

Figure (30) presents the final simulation, showing how the system responds as the sensor resistance R_S decreases from 200 k Ω to 25 k Ω . The graphs illustrate the behavior of the main components: the comparator with hysteresis (bottom), the LED (middle), and the relay (top).

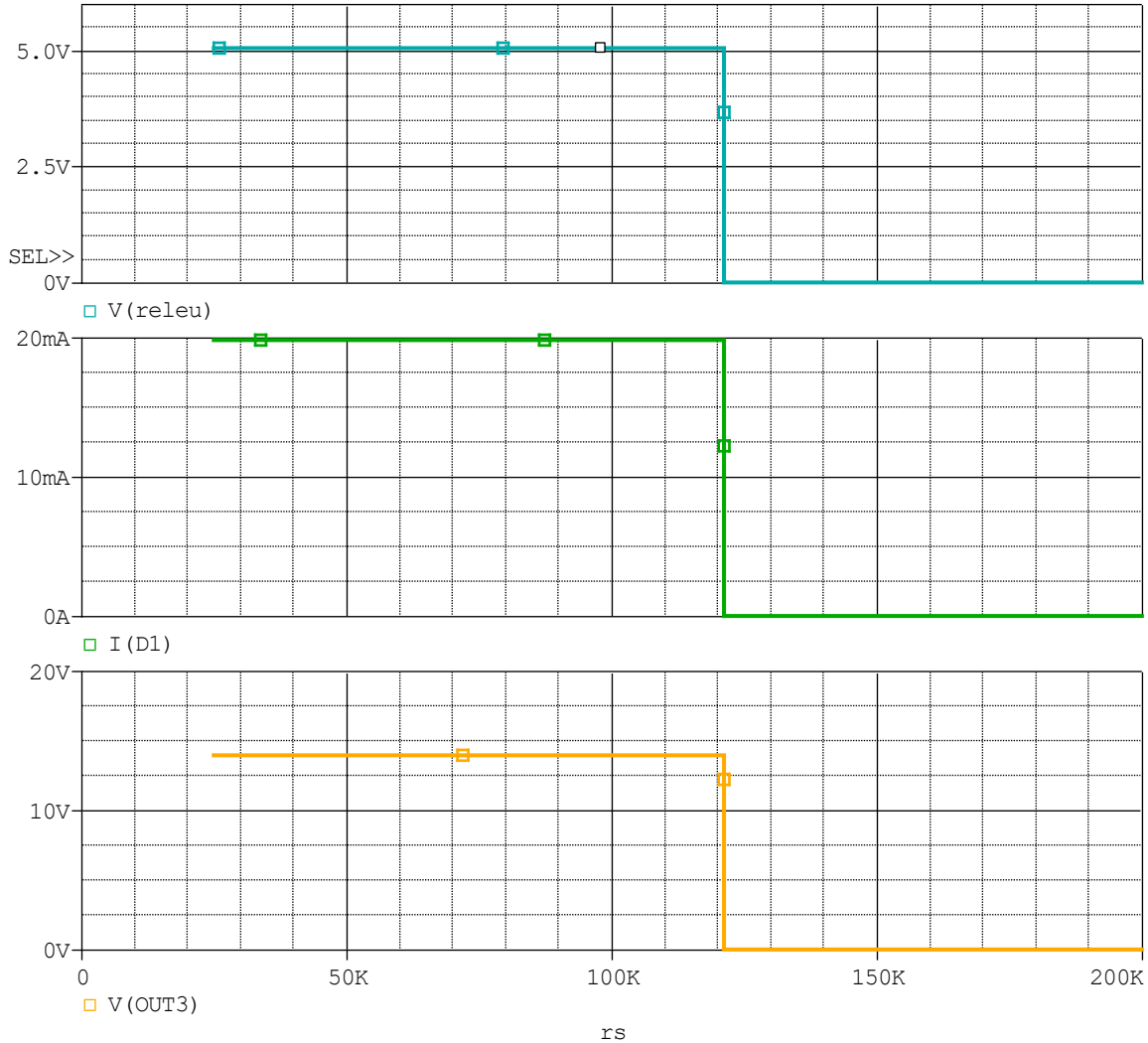


Figure (30). Simulations of Non-Inverting Hysterezis, Orange LED and 5V Relay

As R_S falls below the threshold, the comparator output switches to *LOW*, which turns off both the LED and the relay. This highlights how the circuit components work together based on changes in sensor input.

2.3.2. Worst Case Analysis

Sensitivity identifies which parameters are critical to the circuit operation. This analysis determines to what extent each component affects the circuit operation (single action or with other components). In addition, it changes all the values to simulate the worst case.

A worst-case analysis was performed to evaluate how resistor tolerances affect the switching thresholds of the hysteresis comparator (V_{out3}) and the output range of the voltage domain converter (V_{out2}). For V_{out2} , Figure (31) shows the simulation setup, and Figure (33) displays the results. The purple line correspond to nominal resistor values, while the blue one represent the worst-case scenario within tolerance limits (see the results in Figure (32)).

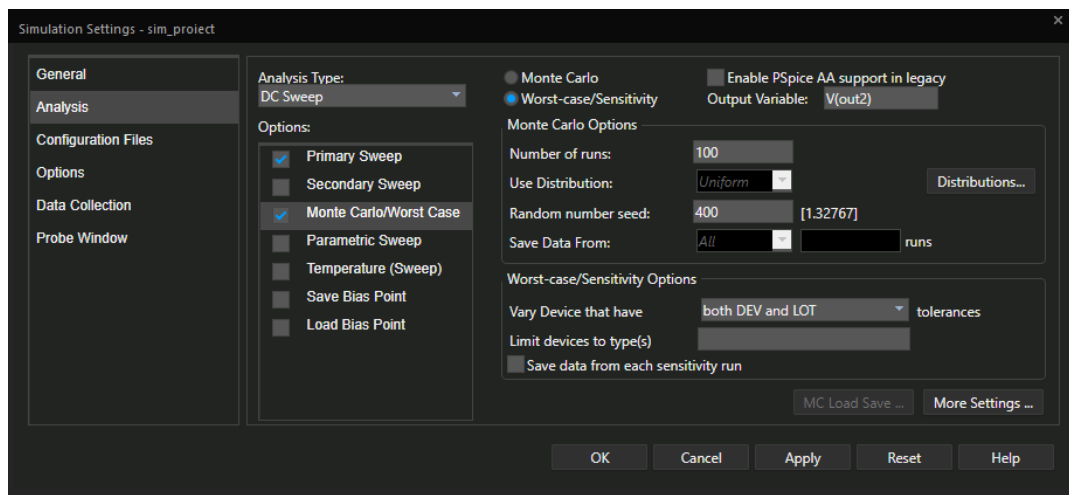


Figure (31). Setting the profile for the Worst Case Analysis

Trace Color	Trace Name	Y1	Y2
	X Values	25.200K	200.000K
CURSOR 1,2	V(out2)	12.002	2.0013
	V(out2)	12.380	2.4618

Figure (32).

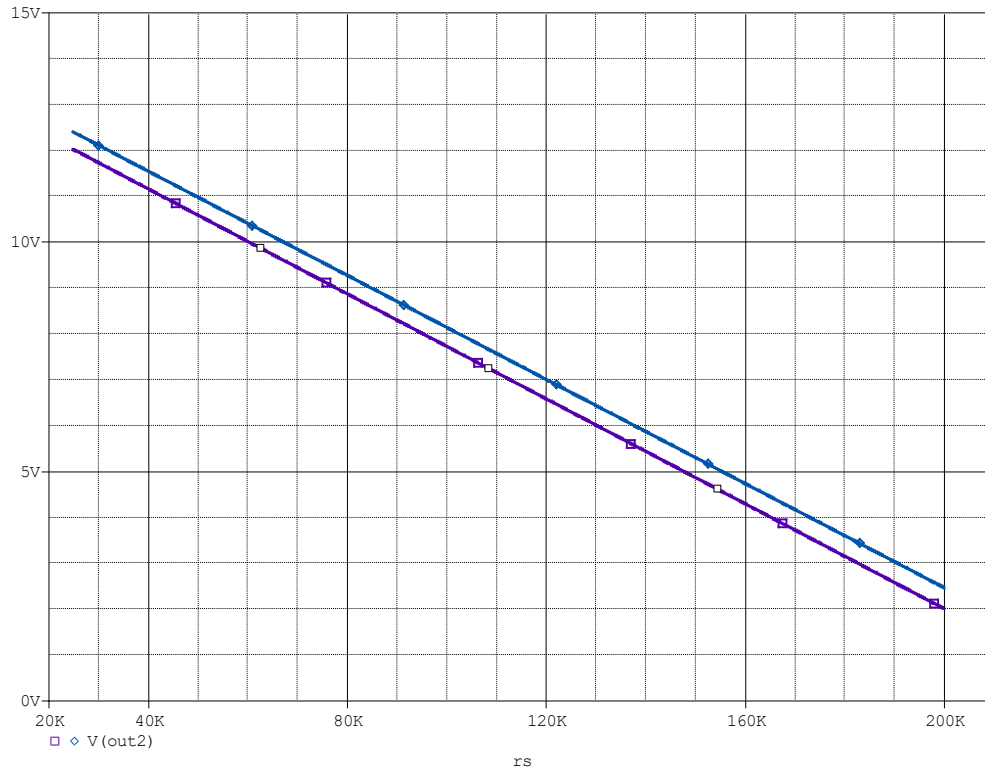


Figure (33).

For V_{out3} , the simulation setup remains, and the results are illustrated in Figure (34). The pink curve corresponds to the nominal resistor values, while the blue curve reflects the worst-case variation caused by resistor tolerances.

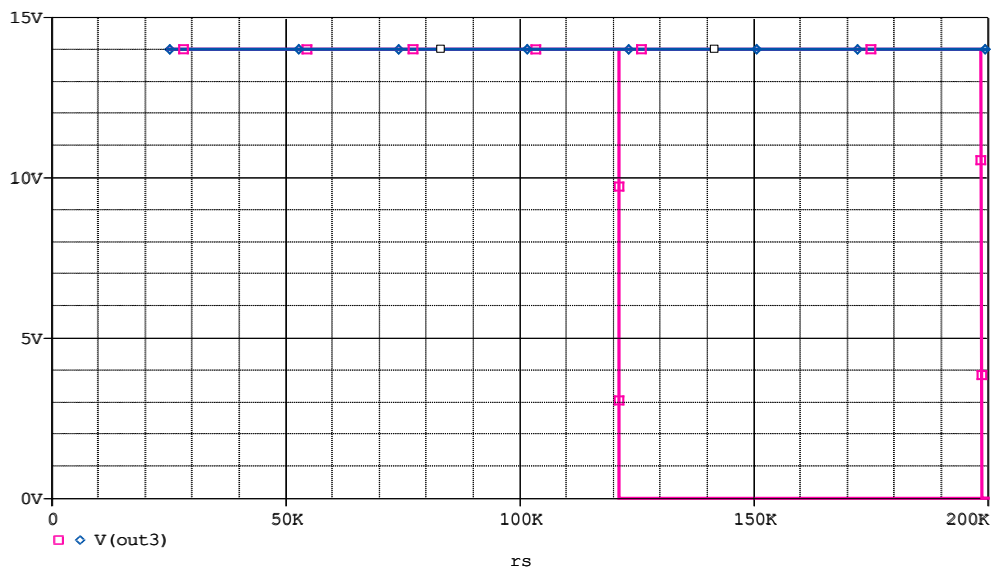


Figure (34).

Device	MODEL	PARAMETER	NEW VALUE
R_R1	R_R1	R	.99 (Decreased)
R_R2	R_R2	R	1.01 (Increased)
R_R3	R_R3	R	1.01 (Increased)
R_R6	R_R6	R	.99 (Decreased)
R_R5	R_R5	R	1.01 (Increased)
R_R7	R_R7	R	1.01 (Increased)
R_R8	R_R8	R	.99 (Decreased)
R_R10	R_R10	R	.99 (Decreased)
R_R9	R_R9	R	1.01 (Increased)
R_R12	R_R12	R	1.01 (Increased)
R_R11	R_R11	R	.99 (Decreased)
R_R13	R_R13	R	1.01 (Increased)
R_R14	R_R14	R	.99 (Decreased)

Figure (35). *LOW* to *HIGH*. Sweeping R_s from 25k Ω to 200k Ω

Device	MODEL	PARAMETER	NEW VALUE
R_R1	R_R1	R	.99 (Decreased)
R_R2	R_R2	R	1.01 (Increased)
R_R3	R_R3	R	1.01 (Increased)
R_R6	R_R6	R	1.01 (Increased)
R_R5	R_R5	R	.99 (Decreased)
R_R7	R_R7	R	1.01 (Increased)
R_R8	R_R8	R	.99 (Decreased)
R_R10	R_R10	R	1.01 (Increased)
R_R9	R_R9	R	.99 (Decreased)
R_R12	R_R12	R	1.01 (Increased)
R_R11	R_R11	R	.99 (Decreased)
R_R13	R_R13	R	1.01 (Increased)
R_R14	R_R14	R	1.01 (Increased)

Figure (36). *HIGH* to *LOW*. Sweeping R_s from 200k Ω to 25k Ω

2.2.3. Monte-Carlo Analysis

Monte Carlo analysis was used to statistically evaluate how component tolerances affect the output voltage range of the voltage domain converter. Figure (37) shows the simulation setup. In Figure (39), the histogram of the maximum V_{out2} values is displayed on one panel, while the corresponding variation in output range is shown on the other. Figure (40) presents the histogram for the minimum V_{out2} values.

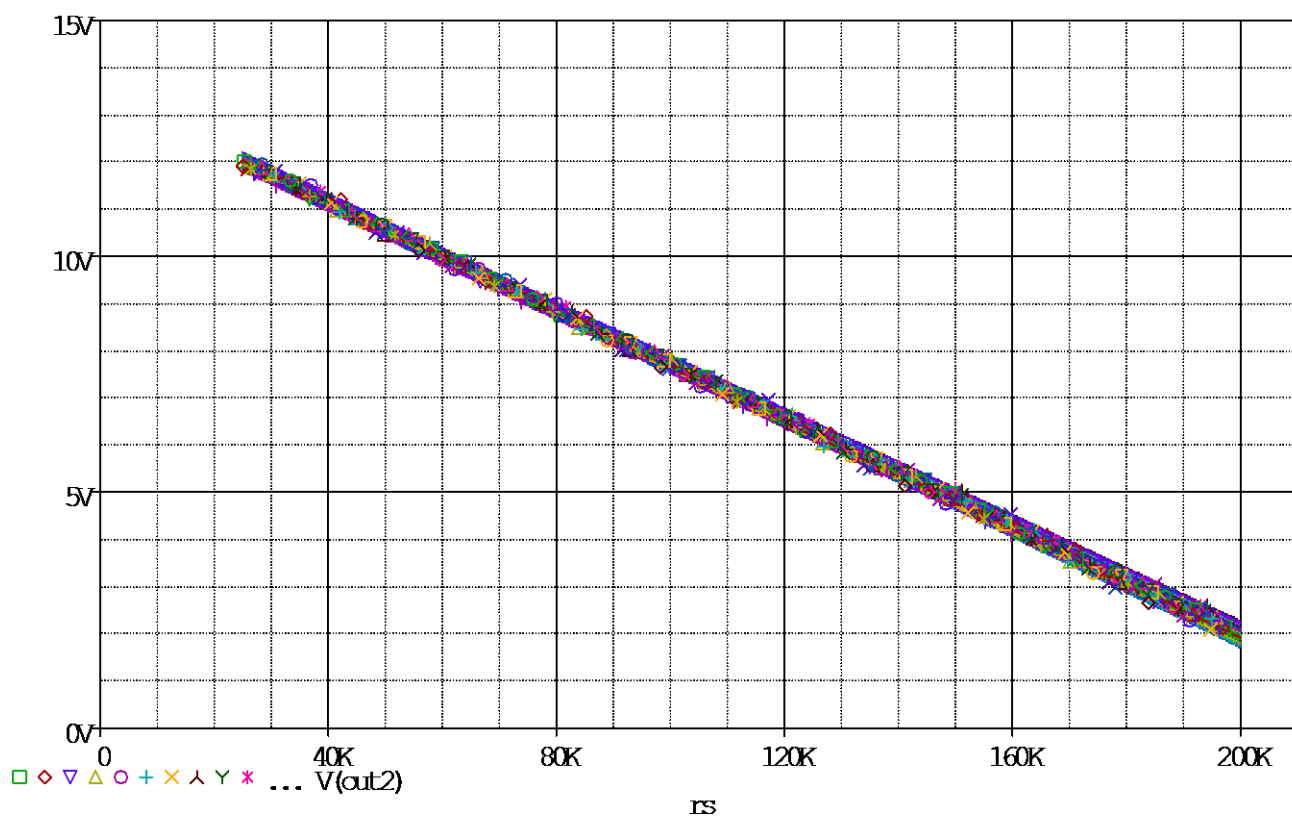


Figure (37).

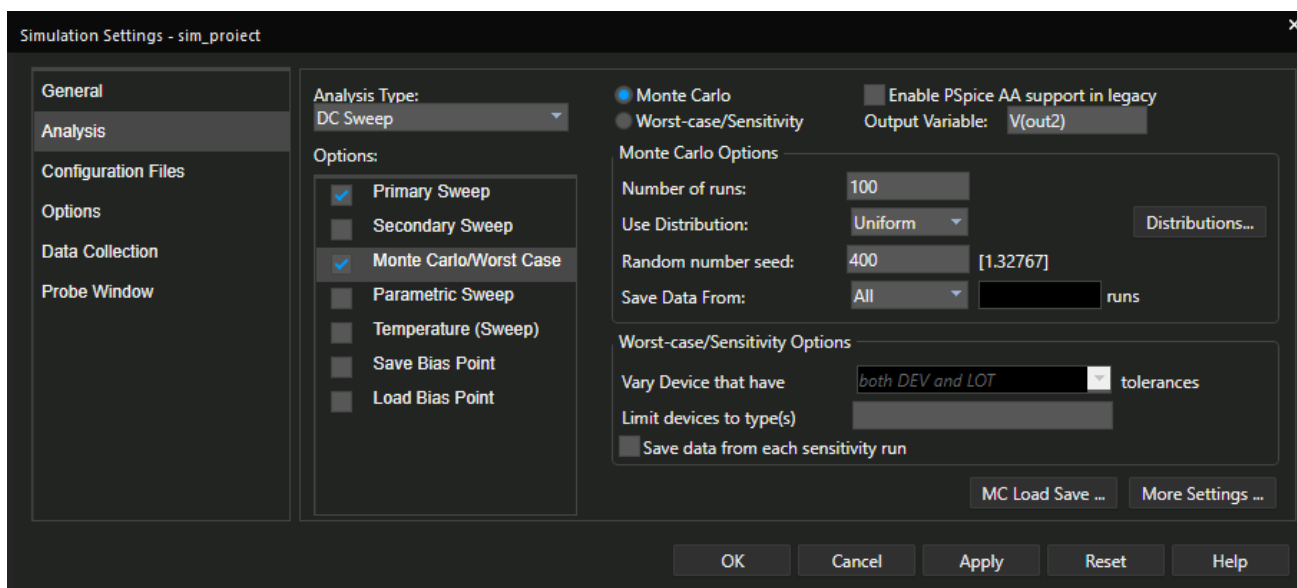


Figure (38). Simulation Profile for Monte Carlo Analysis

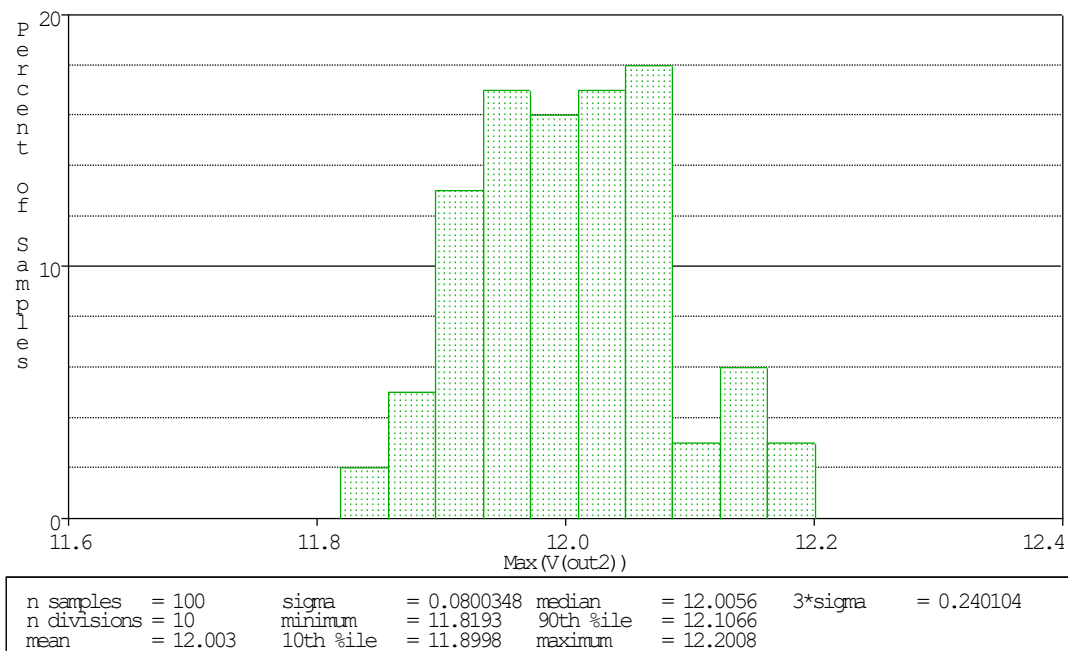


Figure (39).

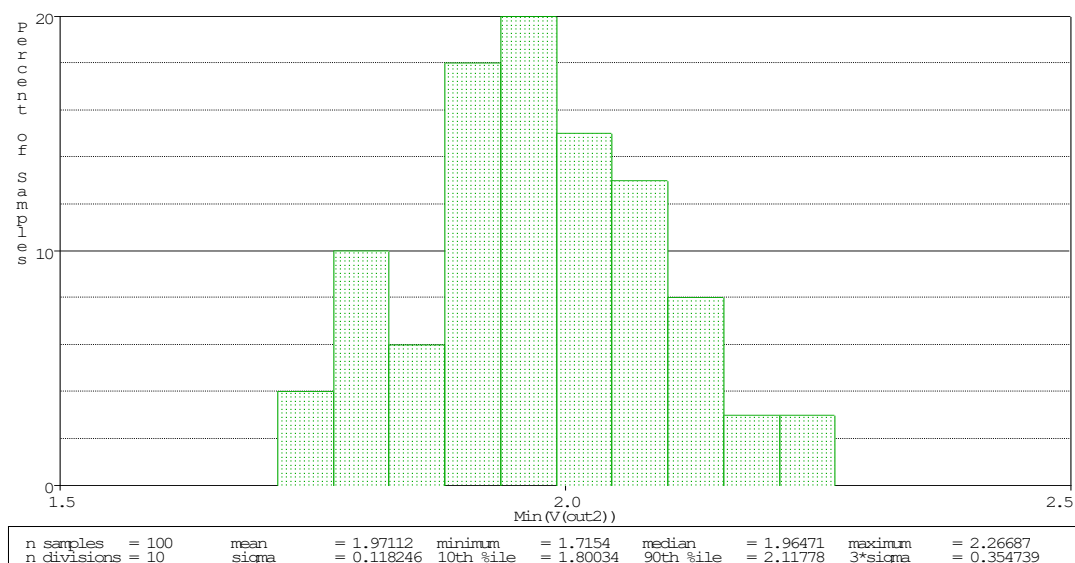


Figure (40).

2.2.4. Temperature Analysis Justification

Temperature analysis was considered unnecessary since the circuit operates indoors, where the ambient temperature typically remains between 15°C and 30°C. Within this range, temperature variations have a negligible impact on component behavior and overall circuit performance.

2.2.5. Complete Circuit Implementation

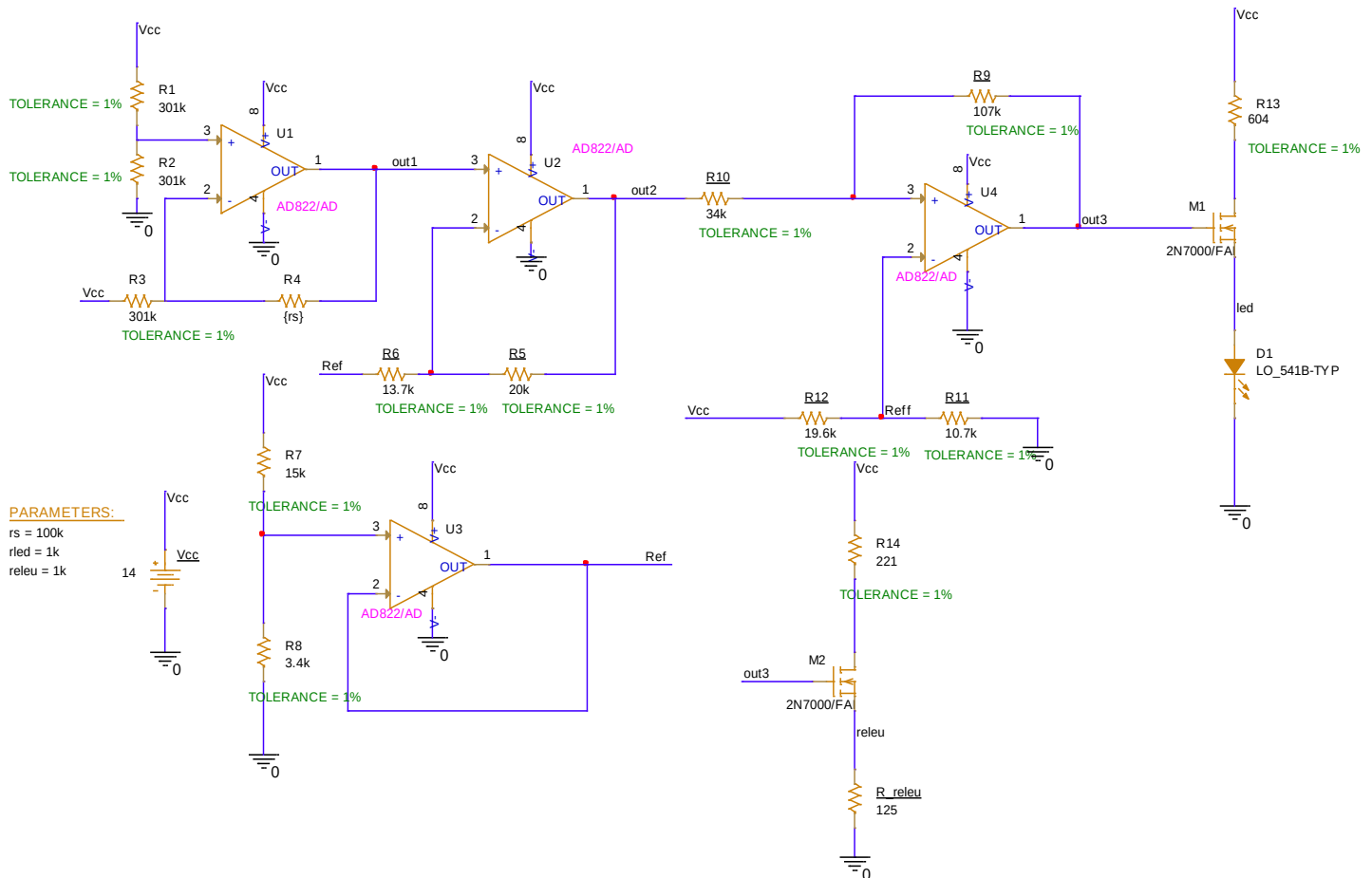


Figure (41). Complete Circuit Schematic

3. Bill Of Materials (BOM)

The BOM was done using the Digi-Key Electronics website ([digikey.com](https://www.digikey.com)) and includes the majority of the components used in the project. The total value of the selected components from Digi-Key is 221.77 RON. Some parts were not available at the time of listing: the LED with part number 475-LA541B-AWDW-24-0-20-BULK-I-ND, estimated at 5.80 RON, and the 125 Ω resistor, estimated at 0.50 RON. These prices were taken into account to provide a complete cost estimate. The detailed component list can be found in Figure (41), and the corresponding Digi-Key BOM is available in the datasheet [7].

+	□	#	PRODUCT	AVAILABILITY	QUANTITY	PACKAGING	UNIT PRICE	EXTENDED PRICE
+	□	1	 MFR-25BFB52-301R  RES 301 OHM 1% 1/4W AXIAL YAGEO	3,961 In Stock View Substitutes	☐ 3 Add Quantity	3 Bulk (301XBK-ND)	0.43000	lei1.29
+	□	2	 MFR-25BFB52-13K7  RES 13.7K OHM 1% 1/4W AXIAL YAGEO	4,200 In Stock View Substitutes	☐ 1 Add Quantity	1 Bulk (13.7KXBK-ND)	0.43000	lei0.43
+	□	3	 MFR-25BFB52-107K  RES 107K OHM 1% 1/4W AXIAL YAGEO	7,073 In Stock View Substitutes	☐ 1 Add Quantity	1 Bulk (107KXBK-ND)	0.43000	lei0.43
+	□	4	 MFR-25BFB52-19K6  RES 19.6K OHM 1% 1/4W AXIAL YAGEO	1,387 In Stock View Substitutes	☐ 1 Add Quantity	1 Bulk (19.6KXBK-ND)	0.43000	lei0.43
+	□	5	 MFR-25BFB52-10K7  RES 10.7K OHM 1% 1/4W AXIAL YAGEO	6,467 In Stock View Substitutes	☐ 1 Add Quantity	1 Bulk (10.7KXBK-ND)	0.43000	lei0.43
+	□	6	 MFR-25BFB52-604R  RES 604 OHM 1% 1/4W AXIAL YAGEO	9,659 In Stock View Substitutes	☐ 1 Add Quantity	1 Bulk (604XBK-ND)	0.43000	lei0.43
+	□	7	 MFR-25BFB52-221R  RES 221 OHM 1% 1/4W AXIAL YAGEO	5,389 In Stock View Substitutes	☐ 1 Add Quantity	1 Bulk (221XBK-ND)	0.43000	lei0.43
+	□	8	 AD822ARZ  IC OPAMP GP 2 CIRCUIT 8SOIC Analog Devices Inc.	0 In Stock View Leadtime View Substitutes	☐ 4 Add Quantity	4 Tube (505-AD822ARZ-ND)	53.39000	lei213.56
+	□	9	 2N7000  MOSFET TO92 N 60V 0.2A 50HM 150C onsemi	36,394 In Stock View Substitutes	☐ 2 Add Quantity	2 Bulk (2N7000FS-ND)	2.17000	lei4.34

Figure (42). List of components from DigiKey

The BOM was also partially completed, shown in Figure (42), using the Mouser Electronics website ([mouser.com](https://www.mouser.com)) and can be found at the address [4]. The total value of the components selected from Mouser is 8.89 RON.

Sort	Product Detail	Description	Quantity	Availability	Unit Price (RON)	Ext. Price (RON)	
1	 Mouser No: 603-MFR-25FBF52-20K Mfr. No: MFR-25FBF52-20K Mfr.: YAGEO Customer No: Customer No	Metal Film Resistors - Through Hole 20K OHM 1/4W 1% RoHS Compliant QuickView	1	1 Dispatches Now	0,47 RON	0,47 RON	
2	 Mouser No: 603-MFR-25FBF52-34K Mfr. No: MFR-25FBF52-34K Mfr.: YAGEO Customer No: Customer No	Metal Film Resistors - Through Hole 34K OHM 1/4W 1% RoHS Compliant QuickView	1	1 Dispatches Now	0,47 RON	0,47 RON	
3	 Mouser No: 660-MF1/4DCT52R1502F Mfr. No: MF1/4DCT52R1502F Mfr.: KOA Speer Customer No: Customer No	Metal Film Resistors - Through Hole 1/4W 15K ohm 1% 15K OHM 1% RoHS Compliant QuickView	1	1 Dispatches Now	0,648 RON	0,65 RON	
4	 Mouser No: 603-MFR-25FBF52-3K4 Mfr. No: MFR-25FBF52-3K4 Mfr.: YAGEO Customer No: Customer No	Metal Film Resistors - Through Hole 3.4K OHM 1/4W 1% RoHS Compliant QuickView	1	1 Dispatches Now	0,47 RON	0,47 RON	
5	 Mouser No: 893-833H-1C-S-5VDC Mfr. No: 833H-1C-S-5VDC Mfr.: Song Chuan Customer No: Customer No	General Purpose Relays GP/IND RELAY RoHS Compliant QuickView	1	1 Dispatches Now	6,83 RON	6,83 RON	
By submitting your order you agree to these terms and conditions . For additional information on availability, click on the Mouser Part No.						Merchandise: 8,89 RON	

Figure (43). List of components from Mouser

The estimated total cost for all components, including the unavailable ones, is 236.96 RON (approximately 47.75 EUR).

4. Conclusion

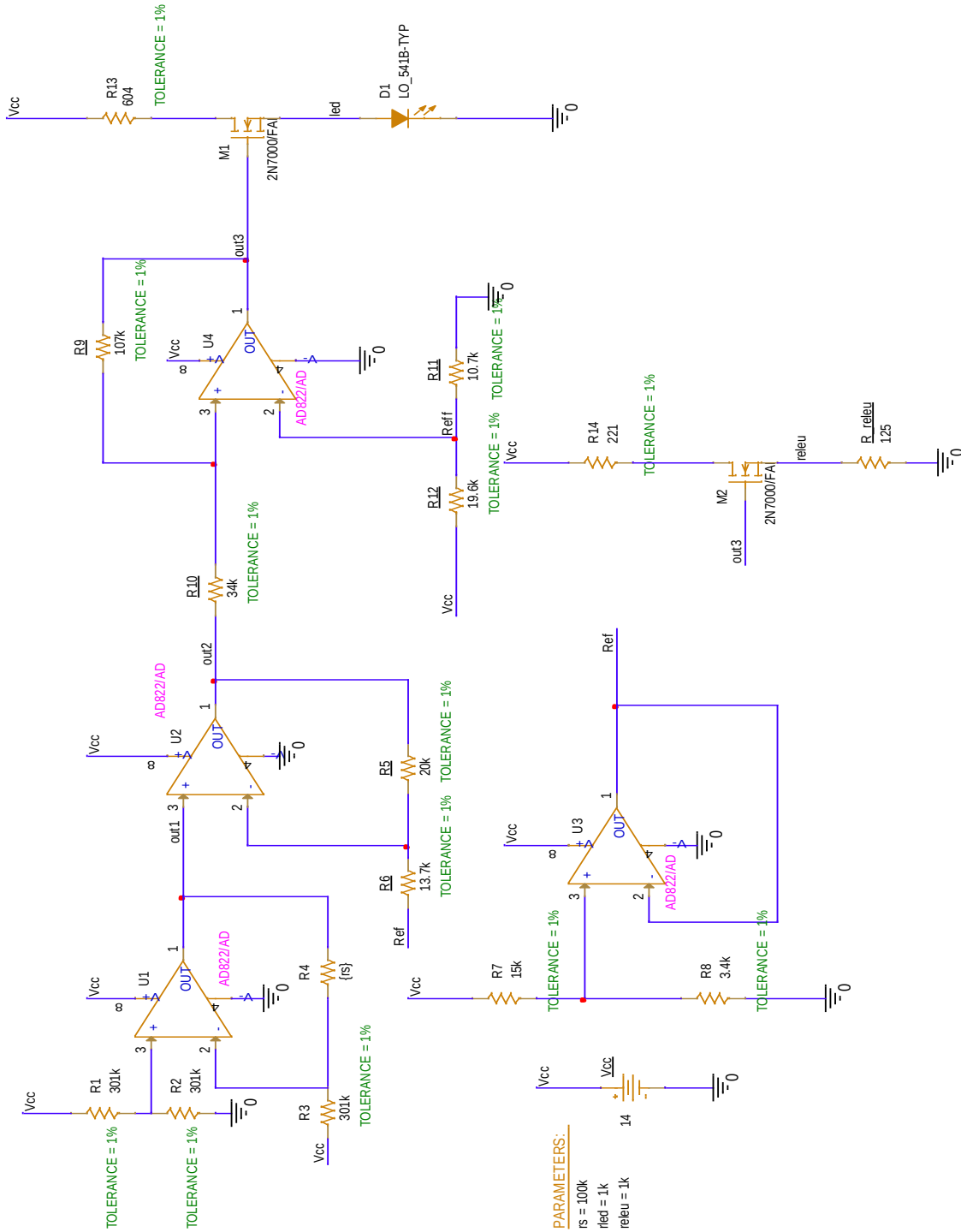
In the end, this project successfully demonstrated the design and simulation of an automatic control circuit for monitoring and regulating carbon monoxide (CO) levels in an enclosed space. The system uses a resistive sensor, signal conditioning stages, and a hysteresis comparator to activate a fan when CO concentrations exceed safe thresholds.

Through simulations, the behavior of each block—active bridge, voltage domain converter, hysteresis comparator, LED, and relay—was verified under various scenarios, including worst-case component tolerances. The system responded reliably to changes in CO concentration, confirming the functionality of the design.

Although minor errors were observed in voltage levels due to resistor tolerances, these can be minimized by selecting components with lower deviation values. The cost of components remains reasonable, even when accounting for unavailable parts.

Overall, the circuit meets its functional requirements and provides a solid foundation for real-world implementation, with potential for further optimization or integration into more complex monitoring systems.

5. Appendix



References

- [1] E. Ș. Laura Ivanciu, *Electronic Devices*, Cluj-Napoca: UTPRESS, 2023.
- [2] L. Ivanciu, 2023, FEC, Available: <http://www.bel.utcluj.ro/dce/didactic/fec/>
- [3] Analog Devices, 2025, AD822 datasheet, Disponibil: <https://www.analog.com/media/en/technical-documentation/data-sheets/AD822.pdf>
- [4] Mouser Electronics, Inc, "Mouser Electronics," [Online]. Available: <https://ro.mouser.com/Cart/>.
- [5] Google AI Studio, 2025, New Chat Prompt, Disponibil: https://aistudio.google.com/prompts/new_chat
- [6] Jansson, "Resistors," [Online]. Available: <https://jansson.us/resistors.html>
- [7] Digi-Key, 2025, PPVQJV94FG, Disponibil: <https://www.digikey.ro/en/mylists/list/PPVQJV94FG>
- [8] RS Components, "LO_541B LED Datasheet," [Online]. Available: <https://docs.rs-online.com/40ab/0900766b808b6aa9.pdf>
- [9] R. F. Ovidiu Pop, *Computer Aided Design*, Cluj-Napoca: UTPRESS, 2016.
- [10] ON Semiconductor, "2N7000 MOSFET Datasheet," [Online]. Available: <https://octopart.com/datasheet/2n7000-onsemi-110503>
- [11] Single Relay, "SRD-05VDC-SL-C Datasheet," [Online]. Available: <https://www.alldatasheet.com/datasheet-pdf/pdf/1132639/SONGLERELAY/SRD-05VDC-SL-C.html>